RGC Ref No. 15225126

RESEARCH GRANTS COUNCIL

General Research Fund

Application Form for 2026/27 (GRF1)

(Please submit your application through RGC Electronic System)

- **Please ensure you read the Scheme Overview and Guidance Notes (Guidance Notes) carefully before completing this form. Section 2 of the Guidance Notes mirrors the application form structure, so will provide you with step-by-step advice on completion.**
- **To safeguard the interests of the researcher and the university, the awardee university bears the primary responsibility for prevention, detection and investigation of research misconduct, including but not limited to misuse of funds, data falsification, plagiarism and double-dipping. The university should vet the grant applications using anti-plagiarism software before submitting them to the RGC.**

PART I: SUMMARY OF THE APPLICATION

[To be completed by the applicant(s)]

1. Particulars of the Project

(a) Name and Academic Affiliation of Principal Investigator (PI):

Name	Post	Unit / Department and University
Prof CHENG, Ran	Associate Professor	Department of Data Science and Artificial Intelligence/The Hong Kong Polytechnic University

(b) Is the PI a new appointee within 2 years of full-time paid appointment to his/her first substantive position as an academic staff in a university at the time of submission of the proposal?

Yes ☐ **No** ☑

(c) Number of hours per week to be spent by the PI in the proposal: 6 hour(s)

(d) Title of Project: Towards Learnable Evolutionary Multi-objective Optimization via Neural Unrolling

(e) Project Duration: 36 Months*

RGC Ref No. 15225126

*** for project with duration exceeding 36 months, please explain in Part II (4) why the proposed research cannot be completed within the normal span of 36 months.**

(f) Total Amount Requested: $ 1,239,756

(g) Nature of Application

New ☑ **Re-submission** ☐ **Continuation** ☐

(h) (i) If this is a re-submission or largely similar to a proposal that has been submitted to but not supported by the UGC/RGC or another funding agency, or a continuation application, please give details here:

Reference No(s). [for UGC/RGC projects only.]:

Name of funding agency(ies) and the funding programme(s):

Project title(s):

Date(month/year) of application:

Outcome:

(ii) If this application is the same as or similar to the one(s) submitted under any UGC / RGC funding schemes but not supported previously, please provide a summary of all the previous submissions, the main concerns / suggestions of the reviewers and the revisions made to each of the previous submissions. Please make sure that sufficient revisions and improvement based on the comments provided by the assessment panels have been made. Maximum of 500 words

N.A.

2. Research Areas and Project Team

(a) Primary Field: Artificial Intelligence and Machine learning **& Code** 2213
Secondary Field: _ **& Code** _

RGC Ref No. 15225126

(b) Project Keywords (maximum of 5):

1) Evolutionary Computation
2) Multi-objective Optimization
3) Evolutionary Algorithm

(c) Project Abstract (comprehensible to a non-specialist)

Evolutionary Multi-objective Optimization (EMO) has long served as a foundational methodology for resolving multiple conflicting objectives in science, engineering, and decision-making. While traditional heuristics have achieved remarkable success in approximating Pareto fronts, the rise of learning-based optimization brings new opportunities for automation, adaptability, and integration with modern AI systems. Yet, the systematic unification of learning and EMO remains an open challenge.

First, conventional EMO algorithms rely on fixed heuristic mechanisms such as non-dominated sorting and diversity preservation. These operators are inherently non-differentiable, creating what we term the Differentiability Gap that hinders end-to-end learning and adaptation. Second, emerging learning-based strategies, including algorithm selection, optimizer composition, and LLM-guided operator design, remain largely confined to single-objective or surrogate-assisted contexts where gradient information is readily available, leaving the problem of Multi-objective Scalability unresolved. Third, real-world scientific problems often demand structural generalization across high-dimensional and computationally intensive environments, yet existing methods lack a principled framework to ensure both adaptability and theoretical soundness, highlighting a persistent Structural Generalization challenge.

To address these challenges, this project proposes a unified framework of Learnable Evolutionary Multi-objective Optimization (LEMO). First, to bridge the Differentiability Gap, we establish a bilevel learning formulation that embeds EMO operations within a differentiable computational graph supported by theoretical guarantees. Second, to enhance Multi-objective Scalability, we design a neural-unrolling architecture equipped with a modular library of learnable operators for efficient optimization in high-dimensional settings. Third, to achieve Structural Generalization, we develop a system-level platform integrating benchmarks, multi-fidelity evaluations, and AI4Science case studies to support validation and deployment of EMO algorithms. Collectively, these efforts aim to advance EMO from a heuristic-driven paradigm toward a learnable, differentiable, and structure-aware framework.

(d) PI and Co-Is

(i) Investigator(s) Information:

Please list the details for the PIs and Co-Is involved in the application in the table (add additional rows if necessary)

Role in the project (e.g. PI, Co-I, etc.)	Name with Title	Post	Unit/ Department/ University	ORCID iD	Current Member of UGC / RGC / Panel / Committee (as at application deadline)? (Yes or No)	Name of UGC / RGC / Panel / Committee
PI	Prof CHENG, Ran	Associate Professor	Department of Data Science and Artificial Intelligence/The Hong Kong Polytechnic University	0000-0001-9410-8263	No	
Co-I(s)	Prof TAN, Kay Chen	Chair Professor	Department of Data Science and Artificial Intelligence/The Hong Kong Polytechnic University	0000-0002-6802-2463	No	
	Prof Yang, Ming	Assistant Professor	Department of Applied Physics/The Hong Kong Polytechnic University	0000-0002-0876-1221	No	
	Dr Yao, Lu	Senior Lecturer	Department of Chemistry/Queen Mary University of London	0000-0001-9566-4122	No	

(ii) CV(s) and Publication List for PI and all Co-Is

(iii) Project team and Plan of Collaboration

1. Composition and Expertise of the Project Team

This project brings together a multidisciplinary team with complementary expertise in evolutionary computation, differentiable learning, and AI-for-Science. The members cover the full spectrum from algorithmic theory and GPU-accelerated implementation to materials and chemistry applications, ensuring that both methodological rigor and scientific relevance are achieved.

(a) Principal Investigator (PI): Dr. Ran Cheng

Associate Professor, Department of Data Science and Artificial Intelligence (DSAI), The Hong Kong Polytechnic University

Dr. Cheng is an internationally recognized researcher in evolutionary computation and learnable optimization, with over 150 publications and multiple IEEE awards (e.g., CIS Outstanding Early Career Award 2025; IEEE TEVC Best Paper Award 2024;). He developed EvoX, a GPU-accelerated platform for evolutionary computation. As PI, he will lead the overall research direction, theoretical formulation (Tasks 1 – 2), and system-level integration (Task 3), while coordinating collaboration among all partners.

(b) Co-Investigator 1: Prof. Kay Chen Tan

Chair Professor & Head, Department of Data Science and Artificial Intelligence, The Hong Kong Polytechnic University

Prof. Tan is a world-leading authority in evolutionary optimization, transfer learning, and intelligent systems, with extensive leadership experience (former Editor-in-Chief, IEEE TEVC; Vice-President, IEEE CIS 2021 – 2024). He will co-lead the algorithmic design of learnable operator libraries and the transfer & generalization strategy (Task 2.4), advising on convergence analysis, scalability, and cross-domain adaptation.

(c) Co-Investigator 2: Dr. Ming Yang

Assistant Professor, Department of Applied Physics, The Hong Kong Polytechnic University

Prof. Yang is an expert in materials physics, 2D materials, and high-throughput computational screening, with over 200 papers in Science, Nature Materials, and Nature Electronics. He will lead Task 3.1 on constructing AI4Science benchmarks based on realistic materials-science datasets, ensuring that the proposed framework is validated on scientifically meaningful, multi-fidelity problems.

(d) Co-Investigator 3: Dr. Yao Lu

Senior Lecturer, Department of Chemistry, Queen Mary University of London (QMUL)

Dr. Lu specializes in materials chemistry and sustainable functional surfaces, with landmark publications in Science and Energy & Environmental Science. She will co-lead benchmark validation and chemical-system case studies under Task 3, providing chemical-domain data and ensuring reproducibility and interpretability of the results.

(e) Supporting Staff: Dr. Jiaxin Gao

Postdoctoral Researcher (no salary requested from GRF), Department of Data Science and Artificial Intelligence, The Hong Kong Polytechnic University

Dr. Gao will assist in algorithmic implementation and experimental coordination, linking theoretical design with GPU-based execution. She will contribute to prototype development, testing, and documentation, ensuring smooth integration among subtasks and reproducible outcomes.

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2. Plan for Collaboration and Coordination

A structured mechanism will ensure efficient coordination and transparent communication.

(a) Core Algorithmic Development (PolyU – DSAI/COMP)

•Lead: Dr. Ran Cheng

•Support: Prof. Kay Chen Tan and Dr. Jiaxin Gao

This subgroup will focus on the unified bilevel formulation, differentiable neural-unrolling architecture, and learnable operator libraries (Tasks 1 & 2). Regular bi-weekly meetings and a shared version-controlled repository will align theoretical advances with implementation progress.

(b) Scientific Benchmarking and Validation (PolyU AP + QMUL)

•Lead: Dr. Ming Yang

•Support: Dr. Yao Lu

This subgroup will construct and validate domain-specific AI4Science benchmarks (Task 3). By combining physics-based simulations with chemical datasets, they will provide meaningful evaluation suites for LEMO. Quarterly coordination meetings will align benchmarks, metrics, and deliverables with algorithmic objectives.

(c) Integration and Dissemination

Dr. Ran Cheng will oversee system integration, supported by Dr. Gao. Annual progress workshops (alternating between Hong Kong and London) will consolidate milestones and promote student exchange. The team will release open-source components and benchmark datasets following PolyU and RGC open-research guidelines.

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3. Synergy and Strategic Value

The team forms a cohesive and complementary structure that links methodological innovation to scientific impact:

•Dr. Ran Cheng & Prof. Kay Chen Tan: Methodological foundation, bilevel optimization, and transferable learning strategies.

•Dr. Ming Yang & Dr. Yao Lu: Domain validation and scientific benchmark construction in materials and chemistry.

•Dr. Jiaxin Gao: Implementation support and experimental coordination ensuring system-level coherence.

This collaboration builds a vertically integrated pipeline (from theory to application) that will establish a learnable, scalable, and scientifically grounded optimization paradigm. The team also serves as a strategic bridge between Hong Kong and the UK, aligning with RGC's mission to advance cross-disciplinary AI-for-Science research.

Dr. CHENG Ran *Senior Member, IEEE*

Academic Qualifications

2013 – 2016: Doctor of Philosophy, University of Surrey, UK.
2006 – 2010: Bachelor of Engineering, Northeastern University, Shenyang, China.

Previous Academic Positions

2016 – 2018: Research Fellow, School of Computer Science, University of Birmingham, UK.
2018 – 2020: Assistant Professor, Department of Computer Science and Engineering, Southern University of Science and Technology, Shenzhen, China.
2021 – 2024: Associate Professor, Department of Computer Science and Engineering, Southern University of Science and Technology, Shenzhen, China.

Present Academic Position

2024 – Present: Associate Professor, Department of Data Science and Artificial Intelligence, Department of Computing, The Hong Kong Polytechnic University, Hong Kong, China.

Relevant Publications in Recent Five Years (*: Corresponding Author)

[1] Jian Yao, **Ran Cheng***, Xingyu Wu, Jibin Wu, and Kay Chen Tan. Diversity-Aware Policy Optimization for Large Language Model Reasoning. In ***Proceedings of the Thirty-ninth Annual Conference on Neural Information Processing Systems (NeurIPS)***, 2025. (**Spotlight**)

[2] Zhenyu Liang, Hao Li, Naiwei Yu, Kebin Sun, and **Ran Cheng***. Bridging Evolutionary Multiobjective Optimization and GPU Acceleration via Tensorization. ***IEEE Transactions on Evolutionary Computation***, 2025. (**IF: 11.7**)

[3] Beichen Huang, **Ran Cheng***, Zhuozhao Li, Yaochu Jin, and Kay Chen Tan. EvoX: A Distributed GPU-accelerated Framework for Scalable Evolutionary Computation. ***IEEE Transactions on Evolutionary Computation***, 2024. (**IF: 11.7**)

[4] Zhichao Lu, **Ran Cheng***, Yaochu Jin, Kay Chen Tan, and Kalyanmoy Deb. Neural Architecture Search as Multiobjective Optimization Benchmarks: Problem Formulation and Performance Assessment. ***IEEE Transactions on Evolutionary Computation***, 28(2), 323-337, 2024. (**IF: 11.7**)

[5] Cheng He, **Ran Cheng***, Lianghao Li, Kay Chen Tan, and Yaochu Jin. Large-scale Multiobjective Optimization via Reformulated Decision Variable Analysis. ***IEEE Transactions on Evolutionary Computation***, 28(1), 47-61, 2024. (**IF: 11.7**)

[6] Hui Bai, **Ran Cheng***, Danial Yazdani, Kay Chen Tan, and Yaochu Jin. Evolutionary Large-scale Dynamic Optimization Using Bi-level Variable Grouping. ***IEEE Transactions on Cybernetics***, 53(11), 6937-6950, 2023. (**IF: 9.4**)

[7] Cheng He, **Ran Cheng***, and Danial Yazdani. Adaptive Offspring Generation for Evolutionary Large-scale Multiobjective Optimization. ***IEEE Transactions on Systems, Man and Cybernetics: Systems***, 52(2), 786-798, 2022. (**IF: 8.6**)

[8] Cheng He, **Ran Cheng***, Ye Tian, Xingyi Zhang, Kay Chen Tan, and Yaochu Jin. Paired Offspring Generation for Constrained Large-Scale Multiobjective Optimization. ***IEEE Transactions on Evolutionary Computation***, 25(3), 448-462, 2021. (**IF: 11.7**)

[9] Cheng He, Shihua Huang, **Ran Cheng***, Kay Chen Tan, and Yaochu Jin. Evolutionary Multiobjective Optimization Driven by Generative Adversarial Networks (GANs). ***IEEE***

Transactions on Cybernetics, 51(6), 3129-3142, 2021. (**IF: 9.4**)

[10] Huangke Chen, **Ran Cheng***, Witold Pedrycz, and Yaochu Jin. Solving Many-objective Optimization Problems via Multistage Evolutionary Search. ***IEEE Transactions on Systems, Man and Cybernetics: Systems***, 51(6), 3552-3564, 2021. (**IF: 8.6**)

Relevant Publications beyond Recent Five Years (*: Corresponding Author)

[1] **Ran Cheng**, Yaochu Jin, Markus Olhofer, and Bernhard Sendhoff. Test Problems forLarge-Scale Multiobjective and Many-Objective Optimization. ***IEEE Transactions on Cybernetics***, 47 (12), 41084121, 2017. (**IF: 9.4**, **Citations: 327**)

[2] **Ran Cheng**, Yaochu Jin, Markus Olhofer, and Bernhard Sendhoff. A Reference Vector Guided Evolutionary Algorithm for Many-Objective Optimization. *IEEE Transactions on Evolutionary Computation*, 20 (5), 773-791, 2016. (**IF: 11.7**, **Citations: 1,514**)

[3] Xingyi Zhang, Ye Tian, **Ran Cheng***, and Yaochu Jin. A Decision Variable Clustering-Based Evolutionary Algorithm for Large-scale Many-objective Optimization. ***IEEE Transactions on Evolutionary Computation***, 22 (1), 97-112, 2016. (**IF: 11.7**, **Citations: 540**)

[4] **Ran Cheng**, and Yaochu Jin. A Competitive Swarm Optimizer for Large scale Optimization. ***IEEE Transactions on Cybernetics***, 45 (2), 191-204, 2015. (**IF: 9.4**, **Citations: 983**)

[5] **Ran Cheng**, Yaochu Jin, Kaname Narukawa, and Bernhard Sendhoff. A Multiobjective Evolutionary Algorithm using Gaussian Process based Inverse Modeling. *IEEE Transactions on Evolutionary Computation*, 19 (6), 838-856, 2015. (**IF: 11.7**, **Citations: 336**)

Honors & Awards

- **Outstanding Early Career Award**, IEEE Computational Intelligence Society, 2025.
- **Best Paper Award**, The 26th Genetic and Evolutionary Computation Conference (GECCO), 2024.
- **Global Highly Cited Researcher**, Clarivate, 2023, 2024.
- **Worldwide Top 2% Scientist**, Stanford University, 2020 – 2024.
- **Outstanding Paper Award**, IEEE Transactions on Evolutionary Computation, 2021.
- **Outstanding Paper Award**, IEEE Computational Intelligence Magazine, 2020.
- **Outstanding PhD Dissertation Award**, IEEE Computational Intelligence Society, 2019.
- **Outstanding Paper Award**, IEEE Transactions on Evolutionary Computation, 2018.

Professional Services

- **Founding Chair**, IEEE Computational Intelligence Society Shenzhen Chapter.
- **Associate Editor**, ACM Transactions on Evolutionary Learning and Optimization.
- **Associate Editor**, IEEE Transactions on Emerging Topics in Computational Intelligence.
- **Associate Editor**, IEEE Transactions on Evolutionary Computation.
- **Associate Editor**, IEEE Transactions on Cognitive and Developmental Systems.
- **Associate Editor**, IEEE Transactions on Artificial Intelligence.
- **Editorial Board Member**, Applied Soft Computing.

TAN Kay Chen (IEEE Fellow), B.Eng. (90 - 94), Ph.D. (94 - 97), University of Glasgow, UK
Chair Professor and Head (2020 - Present), Department of Data Science and AI/The Hong Kong Polytechnic University, Full Professor (2017 - 2020), Department of Computer Science/City University of Hong Kong

Section A: Five Most Representative Publications in Recent Five Years

1. Y. Hu, Z. Huang, R. Liu, X. Xue, X. Sun, L. Song and K. C. Tan, "Source Free Semi-Supervised Transfer Learning for Diagnosis of Mental Disorders on fMRI Scans," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 45, no. 11, pp. 13778-13795, 1 Nov. 2023
2. J. Wu, C. Xu, X. Han, D. Zhou, M. Zhang, H. Li, and K. C. Tan, "Progressive Tandem Learning for Pattern Recognition with Deep Spiking Neural Networks", *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 11, pp. 7824-7840, Nov. 2022.
3. K. C. Tan, L. Feng and M. Jiang, "Evolutionary Transfer Optimization - A New Frontier in Evolutionary Computation Research," *IEEE Computational Intelligence Magazine*, vol. 16, no. 1, pp. 22-33, Feb. 2021.
4. R. Ren, T. Hung and K. C. Tan, "A Generic Deep-Learning-Based Approach for Automated Surface Inspection," *IEEE Transactions on Cybernetics*, vol. 48, no. 3, pp. 929-940, March 2018.
5. Z. -A. Huang, Z. Zhu, C. H. Yau and K. C. Tan, "Identifying Autism Spectrum Disorder from Resting-State fMRI Using Deep Belief Network," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 32, no. 7, pp. 2847-2861, July 2021.

Section B: Five Most Representative Publications beyond the Recent Five Years

1. Q. Yu, H. Tang, K. C. Tan and H. Li, "Rapid Feedforward Computation by Temporal Encoding and Learning With Spiking Neurons," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 24, no. 10, pp. 1539-1552, Oct. 2013.
2. J. Hu, H. Tang, K. C. Tan and H. Li, "How the Brain Formulates Memory: A Spatio-Temporal Model Research Frontier," *IEEE Computational Intelligence Magazine*, vol. 11, no. 2, pp. 56-68, May 2016.
3. C. Zhang, P. Lim, A. K. Qin and K. C. Tan, "Multiobjective Deep Belief Networks Ensemble for Remaining Useful Life Estimation in Prognostics," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 28, no. 10, pp. 2306-2318, Oct. 2017.
4. C. K. Goh and K. C. Tan, "A Competitive-Cooperative Coevolutionary Paradigm for Dynamic Multiobjective Optimization," *IEEE Transactions on Evolutionary Computation*, vol. 13, no. 1, pp. 103-127, Feb. 2009.
5. X. Chen, Y. -S. Ong, M. -H. Lim and K. C. Tan, "A Multi-Facet Survey on Memetic Computation," *IEEE Transactions on Evolutionary Computation*, vol. 15, no. 5, pp. 591-607, Oct. 2011.

Awards and Professional Services:

1. RGC Senior Research Fellow, 2026-2030
2. 2024 and 2019 IEEE Computational Intelligence Magazine Outstanding Paper Award
3. 2020 IEEE Transactions on Cybernetics Andrew P. Sage Best Transactions Paper Award
4. 2016 IEEE Transactions on Neural Networks and Learning Systems Outstanding Paper Award
5. 2012 IEEE CIS Outstanding Early Career Award
6. Vice-President, IEEE Computational Intelligence Society (CIS), 2021 – 2024.

7. Chief Co-Editor, Springer Book Series on Machine Learning: Foundations, Methodologies, and Applications, 2020 – Present
8. Series Editor, Lecture Notes in Electrical Engineering, 2022 – Present
9. Engineering Panel of Joint Research Schemes (JRS) of the Research Grants Council (RGC), 2019 – 2025 (Hong Kong SAR)
10. Editor-in-Chief, IEEE Transactions on Evolutionary Computation, 2015 – 2020.
11. Editor-in-Chief, IEEE Computational Intelligence Magazine, 2010 – 2013.

YANG Ming, Ph. D. | Assistant Professor | Department of Applied Physics
Hong Kong Polytechnic University | Phone: (852) 2766 5665 | Email: mingyang@polyu.edu.hk
ORCID: 0000-0002-0876-1221

Education

- 2005-2010 Ph. D. in Physics, National University of Singapore (NUS), Singapore.
- 2002-2005 Postgraduate study in Optics, Xiamen University, China.
- 1997-2001 B. Sc in Physics, Fujian Normal University, China.

Employment

- 2020-present Assistant Professor, Hong Kong Polytechnic University, Hong Kong.
- 2015-2020 Scientist, Institute of Materials Research and Engineering, Singapore.
- 2009-2015 RA/Postdoctoral Fellow, Department of Physics, NUS, Singapore.

Research Interest

My recent research focuses on accelerating the development of functional materials using high-throughput screening techniques, large-scale DFT calculations, and machine learning. I am also engaged in developing 2D electronic and spintronic devices, as well as exploring the electronic, magnetic, topological, and optical properties of 2D materials and their heterostructures. To date, I have published over 200 peer-reviewed papers in renowned journals, including *Science*, *Nat. Mater., Nat. Electron., Nature Nanotech., PRL, Adv. Mater., and JACS*. My work has received more than 8,800 citations, with an H-index of 49 (as of September 2025, Google Scholar). Additionally, I have contributed to two book chapters and filed three patents.

Publication List A - Five most representative publications in recent five years

1) Polar and quasicrystal vortex observed in twisted-bilayer molybdenum disulfide, Chi Shing Tsang, Xiaodong Zheng, Tong Yang, Zhangyuan Yan, Wei Han, Lok Wing Wong, Haijun Liu, Shan Gao, Ka Ho Leung, Chun-Sing Lee, Shu Ping Lau, **Ming Yang***, Jiong Zhao*, Thuc Hue Ly*, ***Science*** 386, 198-205, 2024.
2) High-throughput identification of exfoliable two-dimensional materials with active basal planes for hydrogen evolution, Tong Yang, Jun Zhou, Ting Ting Song, Lei Shen, Yuan Ping Feng* & **Ming Yang***, ***ACS Energy Letters***7, 2313-2321, 2020
3) Improving carrier mobility in two-dimensional semiconductors with rippled materials, Hong Kuan Ng, Du Xiang, Ady Suwardi, Guangwei Hu, Ke Yang, Yunshan Zhao, Tao Liu, Zhonghan Cao, Huajun Liu, Shisheng Li, Jing Cao, Qiang Zhu, Zhaogang Dong, Chee Kiang Ivan Tan, Dongzhi Chi, Cheng-Wei Qiu, Kedar Hippalgaonkar, Goki Eda, **Ming Yang***, Jing Wu*, ***Nature Electronics*** 5, 489-496, 2022.
4) Phase-controllable large-area two-dimensional In_2Se_3 and ferroelectric heterophase junction, Wei Han, Xiaodong Zheng, Ke Yang, Chi Shing Tsang, Fangyuan Zheng, Lok Wing Wong, Ka Hei Lai, Tiefeng Yang, Qi Wei, Mingjie Li, Weng Fu Io, Feng Guo, Yuan Cai, Ning Wang, Jianhua Hao, Shu Ping Lau, Chun-Sing Lee, Thuc Hue Ly*, **Ming Yang***, Jiong Zhao*, ***Nature Nanotechnology*** 18, 55-63, 2023.
5) Deciphering the ultra-high plasticity in metal monochalcogenides, Lok Wing Wong, Ke Yang, Wei Han, Xiaodong Zheng, Hok Yin Wong, Chi Shing Tsang, Chun-Sing Lee, Shu Ping Lau, Thuc Hue Ly*, **Ming Yang***, Jiong Zhao*, ***Nature Materials***, 23,196-204, 2024.

Publication List B - Five most representative publications beyond the recent five years

1) Field-free spin-orbit torque switching *via* out-of-plane spin-polarization induced by an antiferromagnetic insulator/heavy metal interface, Mengxi Wang, Jun Zhou, Xiaoguang Xu*, Tanzhao Zhang, Zhiqiang Zhu, Zhixian Guo, Yibo Deng, **Ming Yang***, Kangkang Meng, Bin He, Jialiang Li, Guoqiang Yu, Tao Zhu, Ang Li, Xiaodong Han, Yong Jiang*, ***Nature Communications***, 14, 2871, 2023
2) Far out-of-equilibrium spin populations trigger giant spin injection into atomically thin MoS_2, Liang Cheng, Xinbo Wang, Weifeng Yang, Jianwei Chai, **Ming Yang**, Mengji Chen, Yang Wu, Xiaoxuan Chen, Dongzhi Chi, Kuan Eng Johnson Goh, Jian-Xin Zhu, Handong Sun, Shijie Wang, Justin C. W. Song, Marco Battiato, Hyunsoo Yang, Elbert E. M. Chia, ***Nature Physics***, 15, 347-351, 2019.
3) Gate-tunable in-plane ferroelectricity in few-layer SnS, Yang Bao, Peng Song, Yanpeng Liu, Zhihui Chen, Menglong Zhu, Ibrahim Abdelwahab, Jie Su, Wei Fu, Xiao Chi, Wei Yu, Wei Liu, Xiaoxu Zhao, Qing-Hua Xu, **Ming Yang***, & Kian Ping Loh*, ***Nano Letters*** 19, 5109-5117, 2019.
4) Room temperature ferromagnetism of monolayer chromium telluride with perpendicular magnetic anisotropy, Rebekah Chua, Jun Zhou, Xiaojiang Yu, Wei Yu, Jian Gou, Rui Zhu, Lei Zhang, Meizhuang Liu, Mark B. H. Breese, Wei Chen, Kian Ping Loh, Yuan Ping Feng, **Ming Yang***, Yu Li Huang* & Andrew T. S. Wee*, ***Advanced Materials*** 21033604, 2021.
5) Large valley splitting in monolayer WS_2 by proximity coupling to an insulating antiferromagnetic substrate, Lei Xu, **Ming Yang**, Lei Shen, Jun Zhou, Tao Zhu, Yuan Ping Feng, ***Physical. Review. B*** 97, 041405, 2018.

Patents

- A novel sputtering system with central electrode and hollow sputtering cathode for 2D film deposition, Jian Wei Chai, Shi Jie Wang, Dong Zhi Chi, and **Ming Yang**, *PCT/SG2018/050462*, 2018.
- A novel memristor structure based on stacked layers of nanocrystalline TMDCs, Silva Henry Medina, Dongzhi Chi, Jianwei Chai, **Ming Yang**, Shijie Wang, Shiwun Tong, and Carlos Manzano*, PCT/SG2020/050533,* 2020.
- Atomically thin 2D platform for optical applications, Zeng Wang, Jinghua Teng, Shijie Wang and **Ming Yang**, *PCT/SG2020/050356,* 2020.

Book Chapters

- Emerging electronic properties of polymorphic 2D-TMDs, Tong Yang, Zishen Wang, Jiaren Yuan, Jun Zhou and **Ming Yang**, Wiley, 2024.
- Interface between Graphene and High-κ Dielectrics for Graphene Science Handbook: Electrical and Optical Properties, **Ming Yang**, Yuan Ping Feng, and Shi Jie Wang, Taylor & Francis Group, 2015.

Service

- 2022-Present, Secretary, IEEE Hong Kong EDS/SSC Joint Chapter.
- 2024-2025, Executive Committee member and Technical Sub-committees chair, EDTM, Hong Kong.
- 2024-present Associate Editor, IEEE Transactions on Emerging Topics in Computational Intelligence
- 2024-Present, Guest Editor for Special Topic “Surface and Interface of Two-dimensional Materials”, Frontier of Physics.

Dr. Yao Lu | Senior Lecturer, Queen Mary University of London

Degrees and Qualifications

09/2013 – 03/2017 **PhD in Materials Science and Chemistry**
Thesis: "Robust Self-Cleaning Coatings".
Department of Chemistry, University College London, London, UK.
09/2010 – 06/2013 **Master of Engineering**
School of Mechanical Engineering, Dalian University of Technology, Dalian, China.
09/2006 – 06/2010 **Bachelor of Science**
School of Mechanical Engineering, Guangxi University, Nanning, China.

Work Experience

08/2023 – Present **Senior Lecturer**
Department of Chemistry, Queen Mary University of London (QMUL), London, UK.
08/2019 – 07/2023 **Lecturer**
Department of Chemistry, Queen Mary University of London (QMUL), London, UK.
04/2017 – 07/2019 **Postdoctoral Research Associate**
Department of Mechanical Engineering, University College London, London, UK.

Other Appointments and Certifications

04/2025 – Present **Entrepreneurial Lead**, Innovate UK ICURe programme, UK.
04/2025 – Present **Lindau Alumni Reviewer** for the 74th Lindau Nobel Laureate Meeting, Germany.
09/2023 – Present **Tutor Abroad**, recruiting incoming and supporting outgoing exchange students with oversea universities, QMUL Chemistry.
12/2022 – Present **Conference Founder and Chair**, International Meeting on Wetting and Carbon Neutralisation series, UK.
09/2022 – Present **Founding Editor**, **Section Editor-in-Chief: Physical Sciences & Engineering** of *International Journal of Innovation and Entrepreneurship*.
02/2022 – Present **Associate Editor** of *Droplet* (Wiley).
11/2022 – Present **Accreditation Evaluator** of national qualifications, European Qualifications Framework (EQF), EU
11/2021 – Present **Fellow of Higher Education Academy (FHEA)**, UK.
08/2019 – 09/2024 **Member**, Examination Paper Review board, QMUL Chemistry
08/2019 – 09/2023 **Coordinator** of Departmental Seminars and Meetings, QMUL Chemistry.
10/2013 – Present **Invited Reviewer** of 56 international journals, e.g. *Nat. Commun*., *Adv. Mater*., etc.

Memberships

Member: Royal Society of Chemistry (RSC); Materials Research Society (MRS); Center for Cell Dynamics (CCD); International Society of Bionic Engineering (ISBE); Centre for Chemical Research
Associate Member: Association of British Chinese Professors (ABCP)

Key Publications

- **Y. Lu**, S. Sathasivam, J. Song, C. R. Crick, C. J. Carmalt, I. P. Parkin, Robust self-cleaning surfaces that function when exposed to either air or oil, ***Science*** 2015, 347, 1132–1135.

- J. Li, X. Qiao, B. He, Y. Zhang, S. Pal, L. Sun, M. Bilal, Z. Su, X. Gao, J. Briscoe, I. Abrahams, M. Li*, Z. Li*, **Y. Lu***, Biomass-derived functional additive for highly efficient and stable lead halide perovskite solar cells with built-in lead immobilisation, *Energy Environ. Sci*. 2025, 18, 5632-5642.
- L. Sun, Q. Zhao, L. Che, M. Li, X. Leng, Y. Long, **Y. Lu***, Multi-stimuli-responsive weldable bilayer actuator with programmable patterns and 3D shapes, *Adv. Funct. Mater*. 2024, 34, 2311398.
- X. Yang, K. Zhuang, **Y. Lu***, X. Wang*, Creation of topological ultra-slippery surfaces for droplet motion control, *ACS Nano* 2021, 15, 2589–2599. (**Featured on issue supplementary cover**)
- F. Chen, Y. Wang, Y. Tian,* D. Zhang, J. Song,* C. R. Crick, C. J. Carmalt, I. P. Parkin, **Y. Lu***, Robust and durable liquid-repellent surfaces, ***Chem. Soc. Rev.*** 2022, 51, 8476-8583. (Featured on issue outside front cover)

In total I have published 129 peer-reviewed papers, with > 10,000 citations, h-index: 52, i10-index 107. A full list of publications can be found in my google scholar.

Awards (Total: ~£114,090)

04/2023 Champion of the *Low Carbon Technology* sector and the third prize of the global final competition, Shenzhen Innovation & Entrepreneurship International Competition (CNY ¥300,000, ~£36,000).

06/2017 Winner of Bayer Open Science Start-up Pitch at the 67th Lindau Nobel Laureate Meeting dedicated to Chemistry (€10,000).

03/2016 Chinese government award for outstanding self-financed students abroad ($6,000).

09/2013 UCL Oversea Research Scholarship (covers tuition fees of the UCL PhD project ~£21,530/year).

(e) Does this application involve special application? If yes, please tick here and attach the relevant additional document under Appendix 4.

☐ **Clinical Research Fellowship Scheme (please attach the relevant additional document under Appendix 4 and this option is only available for applications under Biology and Medicine Panel)**

☐ **Support for Individual Research (research time) (only available for applications under Humanities and Social Sciences Panel and Business Studies Panel)**

RGC Ref No. 15225126

PART II: DETAILS OF THE RESEARCH PROPOSAL

3. Project Objectives

1. Develop a unified bilevel formulation to address the Differentiability Gap, embedding EMO operations into a differentiable framework with theoretical guarantees.
2. Design a scalable neural-unrolled algorithm with a modular library of learnable operators to tackle Multi-objective Scalability in high-dimensional and computationally intensive settings.
3. Build a system-level platform to enhance Structural Generalization, incorporating standardized benchmarks, multi-fidelity evaluation, and representative AI4Science case studies.
4. Deliver an open-source prototype of the proposed LEMO framework and conduct systematic validation across theory, algorithms, and real-world applications.

Other Information

This proposal is motivated by the need for learnable, scalable, and deployable evolutionary multi-objective optimization (EMO) in high-dimensional AI4Science problems. Conventional EMO depends on static, non-differentiable heuristics that hinder end-to-end training and generalization, creating a ``heuristic ceiling''. Existing learning-to-optimize efforts and surrogate-assisted variants improve isolated operators but still lack a unified, differentiable and theoretically grounded pipeline. Neural Unrolling (NU) offers a principled way to turn iterative search into a trainable computation graph, yet it has not been systematically applied to EMO.

The project addresses three key issues: (i) formulating a coherent bilevel framework (LEMO) that couples set-level performance objectives with Pareto-aware search and supports cooperative/non-cooperative scenarios; (ii) designing a scalable computational method that unrolls KM-based iterations and fuses differentiable numerical steps with learnable evolutionary operators for stable, GPU-parallel execution; and (iii) building a standardized, extensible benchmark platform for AI4Science with latent-space tasks, differentiable constraints, multi-fidelity scheduling, and human-in-the-loop feedback.

Expected outcomes include: (1) new theoretical contributions for learnable EMO with provable stability/convergence; (2) efficient algorithms that transfer across tasks and scale to expensive, high-dimensional problems; and (3) an open-source GPU-accelerated platform, benchmarks, and toolkits to enable rigorous evaluation and practical deployment in scientific and industrial applications.

4. Research Project Statement

(This is to be maximum of 10 A4 sides in Research Project Statement and should include five elements)

(i) Research context, (ii) Research questions, and (iii) Research Methods

Attached 7 pages(s) as follows

(iv) A one-page Project timetable / Gantt Chart showing the research activities

Attached 1 pages(s) as follows

(v) A maximum of two non-text pages of attached diagrams, photos, charts and table etc, if any.

Attached 2 pages(s) as follows

5. Pathways to Impact Statement

Attached 2 pages(s) as follows

6. Reference (a maximum of three pages for references is allowed for listing the publications cited in Section 1-2. All full references should be provided, including all authors of each reference.)

Attached 3 page(s) as follows.

7. Output Dissemination Plan

Target timing of dissemination (quarter/year)	Output type (Journal / Conference / Others)	Name of target journal, conference or other dissemination means
Q2 / 2027	Conference	ACM Genetic and Evolutionary Computation Conference (GECCO)
Q1 / 2028	Journal	IEEE Transactions on Evolutionary Computation
Q2 / 2028	Conference	ACM Genetic and Evolutionary Computation Conference (GECCO)
Q2 / 2029	Conference	ACM Genetic and Evolutionary Computation Conference (GECCO)
Q3 / 2029	Journal	IEEE Transactions on Evolutionary Computation
Q4 / 2029	Journal	IEEE Transactions on Evolutionary Computation

(i) Research Context

(a) Background of Research

Evolutionary Multi-objective Optimization (EMO) has become a fundamental approach for solving problems with multiple conflicting objectives, and has been successfully applied in domains such as engineering design, scheduling, and complex system optimization [1–4]. However, as applications in areas like AI4Science grow increasingly high-dimensional and computationally intensive [5–7], the inherent weaknesses of conventional EMO are exposed. These include reliance on static heuristics, non-learnable procedures, and limited capacity for structural generalization, forming a *heuristic ceiling* that restricts adaptability. Although learning-based optimization has introduced promising ideas such as algorithm selection and optimizer composition, most efforts remain confined to single-objective or surrogate settings. Critically, no unified framework exists to make EMO both learnable and generalizable, highlighting the need for a new paradigm beyond heuristics.

Related Works by Others

1. Learning to Optimize

Learning to Optimize (L2O) represents a paradigm shift that blends the prescriptive rigor of classical optimization with the inductive power of machine learning [8–10]. Instead of hand-crafted update rules, L2O learns optimization strategies directly from data, enabling faster convergence and better adaptation across tasks. Early attempts were largely model-free, e.g., multilayer perceptron [11] or RNN-based optimizers [8] that directly map problem instances to solutions, offering flexibility but limited interpretability and scalability. A more successful path is model-based L2O, where classical iterative schemes are reformulated into learnable Transformer architectures, providing both inductive bias and efficiency [12]. This trend has yielded breakthroughs in meta-based single-objective optimization, where learned optimizers rival or surpass hand-crafted heuristics on unseen black-box tasks [13]. Examples include parameterized mutation and crossover guided by gradient feedback [14], reinforcement-learning controllers for evolutionary scheduling [15], and meta-learning frameworks that train optimizers to adapt across tasks [16]. These methods highlight the feasibility of replacing expert heuristics with adaptive, trainable strategies.

However, existing L2O-based evolutionary approaches remain constrained: 1) Most methods are limited to single-objective small-scale or toy benchmarks, with unclear scalability to large, real-world AI4Science tasks. 2) Learned optimizers often lack theoretical guarantees, making stability and convergence unpredictable. 3) Current L2O integrations treat individual operators in isolation, without a unified framework to learn the entire evolutionary pipeline. These gaps motivate the need for a principled, scalable framework that systematically integrates learnability into multi-objective optimization.

2. Learnable EMO Algorithms

Traditional EMO algorithms, such as NSGA-II [17], MOEA/D [18], and HypE [19], have served as foundational tools across engineering and scientific domains. These methods rely on hand-crafted mechanisms, such as non-dominated sorting, decomposition weights, and indicator-based selection, which are effective yet rigid and non-differentiable. This inherent inflexibility limits their adaptability in dynamic, high-dimensional, or computationally expensive settings [20, 21]. To overcome these limitations, recent studies have begun to introduce learnability into EMO pipelines. Early attempts relied on machine learning surrogates (e.g., Gaussian processes, deep neural networks) to approximate expensive evaluations, thereby reducing computation cost in simulation-driven optimization [22–24]. While useful, such surrogate-assisted approaches preserve the heuristic pipeline of classical EMO and provide only uneven improvements. More recent work has pursued deeper integration by embedding data-driven modules directly into the evolutionary loop: learnable discriminators and generators for reproduction [25], learnable evaluators for function approximation [26], and transfer modules for reusing optimization knowledge across tasks [27]. Reinforcement learning has also been explored

to control operator selection or parameter adaptation dynamically, enabling policies that respond to the state of the search [28, 29]. More recently, Large Language Models (LLMs) have been explored as auxiliary knowledge sources, generating new black-box operators or end-to-end meta heuristics through natural-language prompts [30, 31].

However, current learnable EMO algorithms exhibit unresolved shortcomings: 1) Most works target isolated operators or heuristics, without offering a unified framework to integrate them into fully learnable and theoretically grounded pipelines. 2) They lack a consistent mathematical framework integrating gradient propagation with population-based search, and differentiable extensions are promising but still ad hoc, without systematic modeling of the iterative EMO process. 3) These methods improve adaptability but are often task-specific, lacking scalability to high-dimensional or large-scale optimization problems. These gaps highlight the need for a systematic framework, which unifies bilevel modeling, learnable operators, and neural unrolling into a cohesive paradigm.

3. Neural Unrolling Algorithms

A particularly powerful strand within L2O is Neural Unrolling (NU), which reformulates iterative optimization algorithms into deep network-like computational graphs, enabling end-to-end training of algorithmic parameters [32]. Initially developed in signal processing and imaging, this approach has yielded landmark advances such as LISTA for sparse coding [33], ADMM-Net for compressive sensing [34], and primal-dual networks for medical image reconstruction [35]. By embedding structured iteration into a differentiable framework, NU unites the interpretability of model-based methods with the adaptability of deep learning, achieving promising performance across inverse problems and related domains [36, 37].

However, despite the success of NU in domains such as computer vision, its potential has not yet been systematically explored in the EMO field. Specifically, 1) current learnable EMO methods remain confined to operator-level learning and overlook the opportunity to leverage NU for reformulating the entire evolutionary loop into a differentiable and trainable pipeline; 2) existing differentiable extensions are largely ad hoc, failing to employ NU as a principled mechanism to model the iterative evolutionary process, which restricts their generality and scalability; and 3) unrolling the evolutionary loop through NU offers a unique opportunity to enable gradient-based learning of selection and recombination operators, thereby unifying bilevel modeling, learnable operators, and differentiable unrolling into a cohesive and scalable framework. These gaps highlight the potential of a systematic NU-driven framework that fundamentally advances the design of learnable EMO algorithms.

Related Works by Our Project Team

The project team's background aligns with the core requirements of the proposed LEMO framework, spanning algorithmic foundations, theoretical modeling, computational implementation, and real-world application. For the algorithmic foundation, the LEMO framework is built upon EMO, and the team's prior work includes the development of foundational algorithms for many-objective [38–41] and large-scale multi-objective optimization [42–46], with one key publication in this area accumulating over 1,500 citations [38]. As for the theoretical modeling, the project's central methodology leverages NU and bilevel optimization, and the team has published on the mathematical perspectives and algorithm design for these topics in learning and vision contexts [47, 48]. To ensure computational feasibility, which is critical for the unrolled LEMO architecture, the team brings direct experience from leading the development of EvoX [49], a GPU-accelerated framework for evolutionary computation. Furthermore, the team has developed the tensorized representations for EMO [50] and neuroevolution [51] that are necessary for integrating EMO with modern deep learning backends, which the latter work receiving the ACM GECCO 2024 Best Paper Award. For real-world validation, the PI has applied EMO across multiple domains, including vehicle control [39], error estimation [52], and edge device deployment [53]. Particularly, the PI has collaborated with Co-I Dr. Yao Lu on extending EMO to materials design [54]. Furthermore, Co-Is Dr. Ming Yang and Dr. Yao Lu have established long-standing expertise in materials science [55–59].

(b) Feasibility

Preliminary Results. The feasibility of the proposed research is supported by preliminary experimental verifications from a single-objective instantiation of our framework, termed LESO_{nu}. Corresponding to $m = q = 1$ in our general formulation (Eq. (1) in **Task 1.1**), LESO_{nu} is theoretically grounded in Krasnosel'skiĭ–Mann fixed-point theory [60], which guarantees convergence under the non-expansive operator conditions described in **Task 2.1**. Algorithmically, LESO_{nu} unrolls a composite operator that couples learned evolutionary updates with proxy-gradient steps, enhanced by an elite-mask mechanism that adaptively fuses exploration and refinement. As shown in Figure 1, initial experiments validate the effectiveness of this design. LESO_{nu} achieves: **(a)** robust out-of-distribution convergence on surrogate CoCo-BBOB functions, **(b)** competitive empirical cumulative distribution (ECDF) performance on the large-scale LSGO-1000D suite, and **(c)** promising zero-shot generalization in a realistic unmanned aerial vehicle (UAV) path planning task. Taken together, these results demonstrate that the core principles of LEMO (i.e., embedding learnable operators within a differentiable unrolling architecture) deliver fast convergence, robust scalability, and practical effectiveness even in the single-objective setting, providing encouraging evidence for the feasibility and potential impact of the proposed framework.

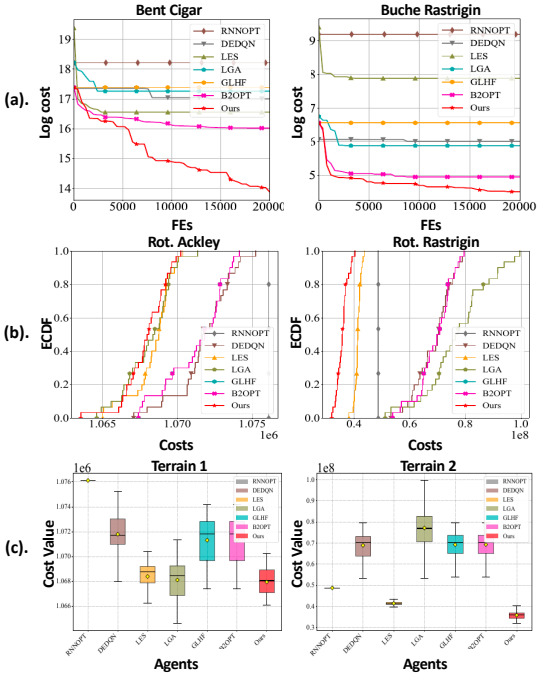


Figure 1: Preliminary results of LESO_{nu} across BBOB, LSGO, and UAV tasks.

(ii) Research Questions

This project is motivated by the need to advance EMO toward methods that are more learnable, scalable, and deployable. To move in this direction, three key research questions are identified.

Q1: How to formulate a unified and theoretically sound framework for LEMO?
Traditional EMO relies on hand-crafted, non-differentiable procedures such as non-dominated sorting and decomposition-based selection, which block gradient flow and hinder adaptability. As a result, these algorithms cannot be trained end-to-end and often require ad hoc hyperparameter tuning for each new problem. Existing learnable variants, such as surrogate-assisted or reinforcement learning strategies, provide partial improvements but operate in isolation and lack consistency across tasks. More critically, most of them do not guarantee Pareto compliance or convergence in both decision and objective spaces, which limits their credibility in scientific applications where theoretical rigor is required. *Therefore, it is essential to develop a coherent bilevel framework that unifies evolutionary and gradient-based operators within a non-expansive unrolling architecture, ensuring learnability, stability, and provable convergence for multi-objective search.*

Q2: How to design a scalable and efficient LEMO algorithm for high-dimensional problems?
Current learnable EMO methods struggle to scale when faced with high-dimensional decision variables or expensive black-box evaluations, as commonly encountered in AI4Science domains such as molecular design or materials discovery. Neural surrogates and operator learning techniques improve certain components, but they remain fragmented, lack structured state-space modeling, and do not exploit parallel hardware effectively. Without efficient gradient propagation or adaptive resource allocation, these methods either fail to converge or become computationally prohibitive. Consequently, they cannot generalize well across high-dimensional problem distributions. *Therefore, a cohesive neural-unrolled architecture is required that integrates modular differentiable operators,*

supports multi-fidelity evaluations, and leverages GPU-parallelized execution to achieve scalable and resource-efficient optimization in complex real-world settings.

Q3: How to build a standardized and extensible benchmark platform for LEMO in AI4Science? The absence of a standardized platform makes it difficult to compare methods, reproduce results, or test applicability in real-world domains. Existing benchmarks for EMO focus mainly on synthetic functions, rarely support latent-space or constraint-aware optimization, and typically lack mechanisms for multi-fidelity evaluation or human-in-the-loop interaction. Moreover, most implementations are CPU-bound and incompatible with differentiable scientific simulators, thereby making them unsuitable for large-scale AI4Science workloads such as drug discovery pipelines or climate modeling. Without such a platform, research progress remains fragmented and difficult to translate into practice. *Therefore, it is necessary to construct a system-level benchmark platform that incorporates structured test suites, scalable GPU execution, and domain-specific modules to enable rigorous validation, cross-domain generalization, and practical deployment of LEMO in scientific and industrial applications.*

(iii) Research Methods

This project establishes the LEMO paradigm through three deeply interlocked tasks, forming a closed-loop research pipeline that spans from mathematical foundations and algorithmic instantiation to system-level validation. Together, they systematically address the core challenges of *Differentiability Gap*, *Multi-objective Scalability*, and *Structural Generalization* in EMO.

Task 1: Unified Formulation

In this task, we establish a unified bilevel optimization framework that bridges evolutionary multi-objective search with gradient-based learning in a differentiable manner. Concretely, we first construct a foundational bilevel formulation that couples set-level performance optimization with Pareto-aware solution exploration, ensuring both solution quality and diversity. We then extend this formulation to handle cooperative and non-cooperative settings, enabling adaptability to diverse decision-making contexts while retaining theoretical consistency and scalability.

Task 1.1: Foundational Bilevel Model

We formulate the basic LEMO model within a bilevel optimization framework, where the upper-level problem aims to maximize the quality of the Pareto front over a distribution of multi-objective problems (MOPs), while the lower-level problem solves a specific MOP to approximate its Pareto front. The quality of the Pareto front is assessed using a set-level performance indicator $\mathcal{L}_{sl}(\cdot)$, such as Hyper-Volume (HV) or Inverted Generational Distance (IGD). Let $\boldsymbol{\omega} \in \mathbb{R}^p$ denote the upper-level decision variable (e.g., the parameters of a neural operator), and let $\mathcal{X} \subset \mathbb{R}^n$ be a compact feasible domain for the lower-level decision variable $\mathbf{x}$. The lower-level objective function is defined as $f(\boldsymbol{\omega}, \mathbf{x}) = (f_1(\boldsymbol{\omega}, \mathbf{x}), f_2(\boldsymbol{\omega}, \mathbf{x}), \ldots, f_q(\boldsymbol{\omega}, \mathbf{x}))^\top$, where q denotes the number of lower-level objectives. The true objectives at the upper-level are represented by $F(\boldsymbol{\omega}, \mathbf{x}) = (F_1(\boldsymbol{\omega}, \mathbf{x}), F_2(\boldsymbol{\omega}, \mathbf{x}), \ldots, F_m(\boldsymbol{\omega}, \mathbf{x}))^\top$, where m is the number of upper-level objectives. The lower-level solution set $\mathcal{S}(\boldsymbol{\omega})$ is defined as the Pareto-optimal set of the lower-level MOP. Let $\mathcal{P}(\boldsymbol{\omega}, \mathbf{x}) = \{F(\boldsymbol{\omega}, \mathbf{x}(\boldsymbol{\omega})) \mid \mathbf{x}(\boldsymbol{\omega}) \in \mathcal{S}(\boldsymbol{\omega})\}$ denote the set of upper-level objective vectors associated with the lower-level Pareto-optimal solutions. The foundational bilevel MOP is formulated as:

$$\min_{\boldsymbol{\omega} \in \mathbb{R}^p, \mathbf{x} \in \mathcal{X}} \mathcal{L}_{sl}\big(\mathcal{P}(\boldsymbol{\omega}, \mathbf{x})\big), \ \text{ s.t. } \mathcal{S}(\boldsymbol{\omega}) = \left\{ \mathbf{x} \in \mathcal{X} \;\middle|\; \nexists\, \mathbf{x}' \in \mathcal{X} \text{ s.t. } f(\boldsymbol{\omega}, \mathbf{x}') \prec f(\boldsymbol{\omega}, \mathbf{x}) \right\}. \tag{1}$$

where $\prec$ denotes Pareto dominance. where $f : \mathbb{R}^p \times \mathbb{R}^n \to \mathbb{R}^q$ defines the auxiliary lower-level optimization objectives, $F : \mathbb{R}^p \times \mathbb{R}^n \to \mathbb{R}^m$ represents the vector-valued upper-level performance objectives, and $\mathcal{L}_{sl} : \mathbb{R}^{n \times m} \to \mathbb{R}$ evaluates the quality of the obtained Pareto set.

Task 1.2: Cooperative Scenario Formulation

In the optimistic scenario, the lower-level problem may admit multiple Pareto-optimal solutions in

the set $\mathcal{S}(\boldsymbol{\omega})$. Here the lower-level is assumed cooperative, selecting the solution most favorable to the upper-level objective. This corresponds to a bilevel minimization problem, where the upper level optimizes over the best-case lower-level response, thus achieving the highest attainable set-level performance $\mathcal{L}_{sl}(\cdot)$ for each $\boldsymbol{\omega}$. In the optimistic case, the formulation of LEMO can be expressed as:

$$\min_{\boldsymbol{\omega}\in\mathbb{R}^p} \left\{ \min_{\mathbf{x}\in\mathcal{S}(\boldsymbol{\omega})} \mathcal{L}_{sl}\big(\mathcal{P}(\boldsymbol{\omega},\mathbf{x})\big) \right\}, \quad \text{s.t.} \quad \mathcal{S}(\boldsymbol{\omega}) = \left\{ \mathbf{x}\in\mathcal{X} \;\middle|\; \nexists\, \mathbf{x}'\in\mathcal{X} \text{ s.t. } f(\boldsymbol{\omega},\mathbf{x}') \prec f(\boldsymbol{\omega},\mathbf{x}) \right\}. \tag{2}$$

Task 1.3: Non-Cooperative Scenario Formulation

In the pessimistic scenario, the lower-level problem may also admit multiple Pareto-optimal solutions in $\mathcal{S}(\boldsymbol{\omega})$, but the lower-level is assumed adversarial, selecting the solution least favorable to the upper-level objective. This corresponds to a min–max bilevel problem, where the upper level optimizes under the worst-case lower-level response. The pessimistic formulation of LEMO is given by:

$$\min_{\boldsymbol{\omega}\in\mathbb{R}^p} \left\{ \max_{\mathbf{x}\in\mathcal{S}(\boldsymbol{\omega})} \mathcal{L}_{sl}\big(\mathcal{P}(\boldsymbol{\omega},\mathbf{x})\big) \right\}, \quad \text{s.t.} \quad \mathcal{S}(\boldsymbol{\omega}) = \left\{ \mathbf{x}\in\mathcal{X} \;\middle|\; \nexists\, \mathbf{x}'\in\mathcal{X} \text{ s.t. } f(\boldsymbol{\omega},\mathbf{x}') \prec f(\boldsymbol{\omega},\mathbf{x}) \right\}. \tag{3}$$

Task 2: Computational Method

In this task, we develop a scalable LEMO algorithm for efficient high-dimensional multi-objective optimization under a unified bilevel learning paradigm. We first design a neural unrolling architecture based on Krasnosel'skiĭ–Mann theory to ensure stable convergence in end-to-end training. Then, we build a modular operator library combining differentiable numerical steps with learnable evolutionary operatoes, enabling flexible hybrid strategies and adaptive search across diverse problem scales.

Task 2.1: Differentiable Neural Unrolling Architecture

In bilevel multi-objective optimization, the lower-level problem, parameterized by the upper-level variable $\boldsymbol{\omega}$, aims to approximate the Pareto-optimal set $\mathcal{S}(\boldsymbol{\omega})$. To preserve solution diversity and ensure stable convergence while supporting end-to-end learning, we adopt a neural unrolling view of the solver, as shown in Figure 4(a). The update is based on α-averaged non-expansive Krasnosel'skiĭ–Mann (KM) iterations[60], integrating a composite operator:

$$\mathbf{x}^{k+1} := \mathcal{T}_{NU}(\boldsymbol{\omega},\mathbf{x}^k) = \mathbf{x}^k + \alpha\left(\mathcal{NU}_{\text{MO}}(\mathbf{x}^k;\boldsymbol{\omega},\mathbf{w}) - \mathbf{x}^k\right),\ \mathcal{NU}_{\text{MO}}(\cdot;\boldsymbol{\omega},\mathbf{w}) \in \{\mathcal{NU}_{\text{evo}}(\cdot;\boldsymbol{\omega},\mathbf{w}) \circ \mathcal{NU}_{\text{num}}(\cdot;\boldsymbol{\omega},\mathbf{w})\}. \tag{4}$$

Here, $\mathcal{T}_{\text{NU}}(\cdot)$ denotes the KM-based mapping, $\alpha \in (0,1)$ balances the current solution and update, and "$\circ$" indicates composition. The composite operator $\mathcal{NU}_{\text{MO}}$ couples: (i) $\mathcal{NU}_{\text{num}}$, an unrolled numerical MOO step (e.g., scalarization descent, decomposition updates, or Pareto-selection); (ii) $\mathcal{NU}_{\text{evo}}$, a learnable evolutionary refinement parameterized by $\boldsymbol{\omega}$. A weight vector $\mathbf{w} \in \mathbb{R}^q$, sampled from distribution $\mathcal{W}$, specifies the objective trade-off and is shared across both modules to ensure consistent preferences along the unrolled pipeline. With scalarization, $\phi_{\mathbf{w}}$ (e.g., weighted sum, Tchebycheff, or R2) maps objectives $f(\boldsymbol{\omega},\mathbf{x})$ to a scalar descent direction. Sampling diverse $\mathbf{w}$ across steps enhances Pareto front coverage. The α-averaged KM wrapper mitigates oscillations and preserves the non-expansive structure necessary for stable unrolling and backpropagation through all K layers.

Task 2.2: Differentiable Operator Library

To enhance the flexibility and adaptability of the proposed LEMO via the NU framework, we will construct a *modular library of differentiable operators*, as illustrated in Figure 4(b). This library supports plug-and-play customization for both the numerical component $\mathcal{NU}_{\text{num}}$ and the evolutionary component $\mathcal{NU}_{\text{evo}}$ within the composite update $\mathcal{NU}_{\text{MO}}$ in Eq. (4). In this way, the unrolled optimization process can be readily adapted to diverse classes of multi-objective problems while remaining fully differentiable.

1) **Numerical Operator Library** ($\mathcal{NU}_{\text{num}}$). This library provides differentiable variants of classical multi-objective optimization steps, enabling gradient backpropagation across all K unrolled layers. Representative implementations include: (i) scalarization-based gradient descent (e.g., weighted sum, Tchebycheff, R2 indicator); (ii) decomposition-based updates as in MOEA/D, where each subproblem

is associated with a weight vector $\mathbf{w}$; (iii) smooth Pareto-dominance selection using differentiable dominance scores as a surrogate for non-dominated sorting.

2) **Evolutionary Operator Library** ($\mathcal{NU}_{\text{evo}}$). This library introduces learnable neural operators that refine the numerical update $\mathcal{NU}_{\text{num}}(\cdot)$ by capturing complex heuristics and adaptive search patterns from data. Specifically, $\mathcal{NU}_{\text{evo}}$ incorporates three canonical paradigms in EMO: (i) dominance-based operators that learn a differentiable pairwise score s_{ij} as a "soft" Pareto dominance; (ii) decomposition-based operators with learnable weights λ that dynamically adjust scalarization functions; (iii) indicator-driven operators that approximate set-level indicators (e.g., HV, IGD) with neural surrogates and back-propagate their gradients $\nabla_{\boldsymbol{\omega}}\mathcal{L}_{sl}$ to guide solution sets.

Task 2.3: Bilevel Training Algorithm

As illustrated in Figure 4(c), the proposed LEMO framework is trained in a bilevel manner, where the lower-level problem optimizes decision variables $\mathbf{x}$ for a fixed upper-level parameter $\boldsymbol{\omega}$, and the upper-level problem updates $\boldsymbol{\omega}$ according to the set-level quality of the resulting Pareto front. This task formalizes the coupled optimization process and the loss functions for end-to-end training.

1) **Lower-Level Update.** At each outer iteration t, $\mathbf{x}$ is refined through K unrolled steps by sequentially applying: (i) the unrolled numerical operator $\mathcal{NU}_{\text{num}}$ (Eq. (4)); (ii) the evolutionary refinement $\mathcal{NU}_{\text{evo}}$ from Task 2.2; and (iii) the α-averaged KM iteration for stability. The resulting direction $\mathbf{x}^k_{\text{NU}}$ is fused with the upper-level indicator direction $\mathbf{v}^k_u$ via an adaptive aggregation rule, where the fusion weight μ^k is determined by cosine similarity:

$$\mu^k = \sigma\left(\beta \cdot \frac{\langle \mathbf{v}^k_u, \mathbf{x}^k_{\text{NU}} \rangle}{\|\mathbf{v}^k_u\| \, \|\mathbf{x}^k_{\text{NU}}\|}\right), \quad \mathbf{x}^k \leftarrow \text{Proj}_{\mathcal{X}}\big(\mu^k \mathbf{v}^k_u + (1-\mu^k)\mathbf{x}^k_{\text{NU}}\big), \tag{5}$$

where $\beta > 0$ controls sensitivity. This balances global guidance with local search: high similarity promotes blending, while conflicting signals bias the update toward the more reliable component.

2) **Upper-Level Update.** To enhance stability and efficiency, the outer update of $\boldsymbol{\omega}$ adopts an extragradient step with a natural hypergradient preconditioner, leveraging the Jacobian of the unrolled mapping to match problem geometry:

$$\mathbf{g}^t = \nabla_{\boldsymbol{\omega}}\mathcal{L}_{sl}(\mathcal{P}(\boldsymbol{\omega}^t)), \quad \tilde{\boldsymbol{\omega}} = \boldsymbol{\omega}^t - \gamma(\mathbf{H}^t)^{-1}\mathbf{g}^t, \quad \boldsymbol{\omega}^{t+1} = \boldsymbol{\omega}^t - \gamma(\mathbf{H}^t)^{-1}\nabla_{\boldsymbol{\omega}}\mathcal{L}_{sl}(\mathcal{P}(\tilde{\boldsymbol{\omega}})), \tag{6}$$

where $\mathbf{H}^t \approx (\mathbf{J}_x\mathcal{T}_{\text{NU}})^\top(\mathbf{J}_x\mathcal{T}_{\text{NU}}) + \lambda\mathbf{I}$ is a positive-definite preconditioner estimated from Hessian–vector products, and $(\mathbf{H}^t)^{-1}\mathbf{v}$ is approximated via conjugate gradient or Neumann series. This predictor–corrector scheme mitigates stale gradients, adapts to curvature, and improves convergence in complex bilevel landscapes.

3) **Loss Function Design.** The set-level indicator loss $\mathcal{L}_{sl}$ converts the quality of the Pareto set into a scalar optimization objective for the upper level. Supported differentiable forms include: (i) Negative HyperVolume (HV): $\mathcal{L}_{sl} = -\text{HV}(\mathcal{P}(\boldsymbol{\omega}); \mathbf{r})$, with $\mathbf{r}$ a dominated reference point; (ii) Inverted Generational Distance (IGD): $\mathcal{L}_{sl} = \frac{1}{|P^*|}\sum_{\mathbf{y}\in P^*} \min_{\mathbf{z}\in\mathcal{P}(\boldsymbol{\omega})} \|\mathbf{y}-\mathbf{z}\|_2$, smoothed for differentiability; (iii) Weighted Tchebycheff aggregation: $\mathcal{L}_{sl} = \mathbb{E}_{\mathbf{w}\sim\mathcal{W}}\big[\min_{\mathbf{z}\in\mathcal{P}(\boldsymbol{\omega})} \max_i w_i|z_i - z_i^{\text{ref}}|\big]$. These interchangeable losses enable gradients to backpropagate through the entire unrolled computation graph, unifying numerical updates, evolutionary refinements, and performance-driven adaptation.

Task 2.4: Transfer & Generalization Strategy

To enhance LEMO's practical generalization, we design a two-stage training strategy that ensures both cross-task transfer and rapid adaptation to new problems. In the *offline pre-training* stage, the optimizer (including $\mathcal{NU}_{\text{num}}$ and $\mathcal{NU}_{\text{evo}}$) is trained on a diverse set of proxy MOPs, capturing general optimization patterns and transferable heuristics across heterogeneous domains (Figure 4(d)). In the *online adaptation* stage, meta-learning techniques (e.g., gradient-based adaptation or recurrent meta-optimizers) fine-tune $\boldsymbol{\omega}$ on unseen target problems using limited real-time feedback. This "structure pretraining + rapid adaptation" pipeline allows the optimizer to exploit priors from past trajectories while maintaining the flexibility to quickly specialize to new tasks, thereby achieving both high initial performance and consistent asymptotic convergence in challenging bilevel multi-objective landscapes.

Task 3: Benchmark Platform

In this task, we develop an open and extensible platform that enables systematic validation, efficient execution, and real-world deployment of LEMO, with a focus on AI4Science applications. The platform is composed of three tightly connected components: (i) scientifically grounded benchmarks with latent-space optimization and differentiable constraints, (ii) a GPU-accelerated execution engine with multi-fidelity scheduling, and (iii) an interactive human-in-the-loop mechanism for guided scientific discovery.

Task 3.1: Scientific Benchmark Construction

We transform static scientific datasets (e.g., MoleculeNet, QM9) into dynamic benchmarks directly compatible with the bilevel framework. The pipeline consists of two steps: (i) mapping raw scientific objects into a continuous latent decision space with invertible decoding to ensure feasibility, and (ii) constructing differentiable scientific objectives and set-level losses. Given a raw object $\mathbf{o}$ (e.g., a molecule or crystal), a generative model (e.g., VAE, GNN-VAE, diffusion models) encodes it into latent space $\mathbf{x} = \Phi_{\text{enc}}(\mathbf{o}), \quad \hat{\mathbf{o}} = \Phi_{\text{dec}}(\mathbf{x})$, with Φ_{enc} and Φ_{dec} as invertible mappings. For m scientific objectives $\{M_j(\cdot)\}_{j=1}^m$, surrogate models g_θ^j (Gaussian processes or neural networks) approximate them, yielding $F_j(\boldsymbol{\omega}, \mathbf{x}) := \mu_{\theta^*,j}(\mathbf{x})$ (optionally: $-\kappa\, \sigma_{\theta^*,j}(\mathbf{x})$), where θ^* are trained parameters. Differentiable approximations of set-level indicators are then applied:

$$\mathcal{L}_{sl}(\mathcal{P}(\boldsymbol{\omega})) = -\widetilde{\text{HV}}_\tau(\mathcal{P}(\boldsymbol{\omega}); \mathbf{r}) \quad \text{or} \quad \widetilde{\text{IGD}}_\tau(\mathcal{P}(\boldsymbol{\omega}); P^*), \tag{7}$$

with gradients propagated end-to-end through the unrolled pipeline, ensuring compatibility with both $\mathcal{NU}_{\text{num}}$ and $\mathcal{NU}_{\text{evo}}$.

Task 3.2: GPU-Accelerated Execution

To achieve high-throughput and scalable execution, we design a unified tensorized interface for populations and operators, enabling seamless switching between numerical and evolutionary modules on GPU. A population is represented as $\mathbf{X} \in \mathbb{R}^{B \times n}$, with operator mappings:

$$\mathcal{O}_{\text{num}} : (\mathbf{X}, \boldsymbol{\omega}, \mathbf{W}) \mapsto \mathbf{V}_{\text{num}}, \quad \mathcal{O}_{\text{evo}} : (\mathbf{V}_l, \texttt{num}, \boldsymbol{\omega}, \mathbf{W}, \mathcal{H}) \mapsto \mathbf{V}_l, \tag{8}$$

where $\mathcal{H}$ encodes historical trajectories (e.g., RNN, Transformer). Vectorization is implemented using `vmap`/`jit` in `JAX`/`XLA` or `CUDA Graphs`, enabling kernel fusion and compilation-level optimization. This supports batched KM-unrolling updates:

$$\mathbf{X}^{k+1} = \mathbf{X}^k + \alpha(\mathbf{V}^{lk} - \mathbf{X}^k), \quad \mathbf{V}^{lk} = \mathcal{O}_{\text{evo}}(\mathcal{O}_{\text{num}}(\mathbf{X}^k, \boldsymbol{\omega}, \mathbf{W}); \boldsymbol{\omega}, \mathbf{W}, \mathcal{H}), \tag{9}$$

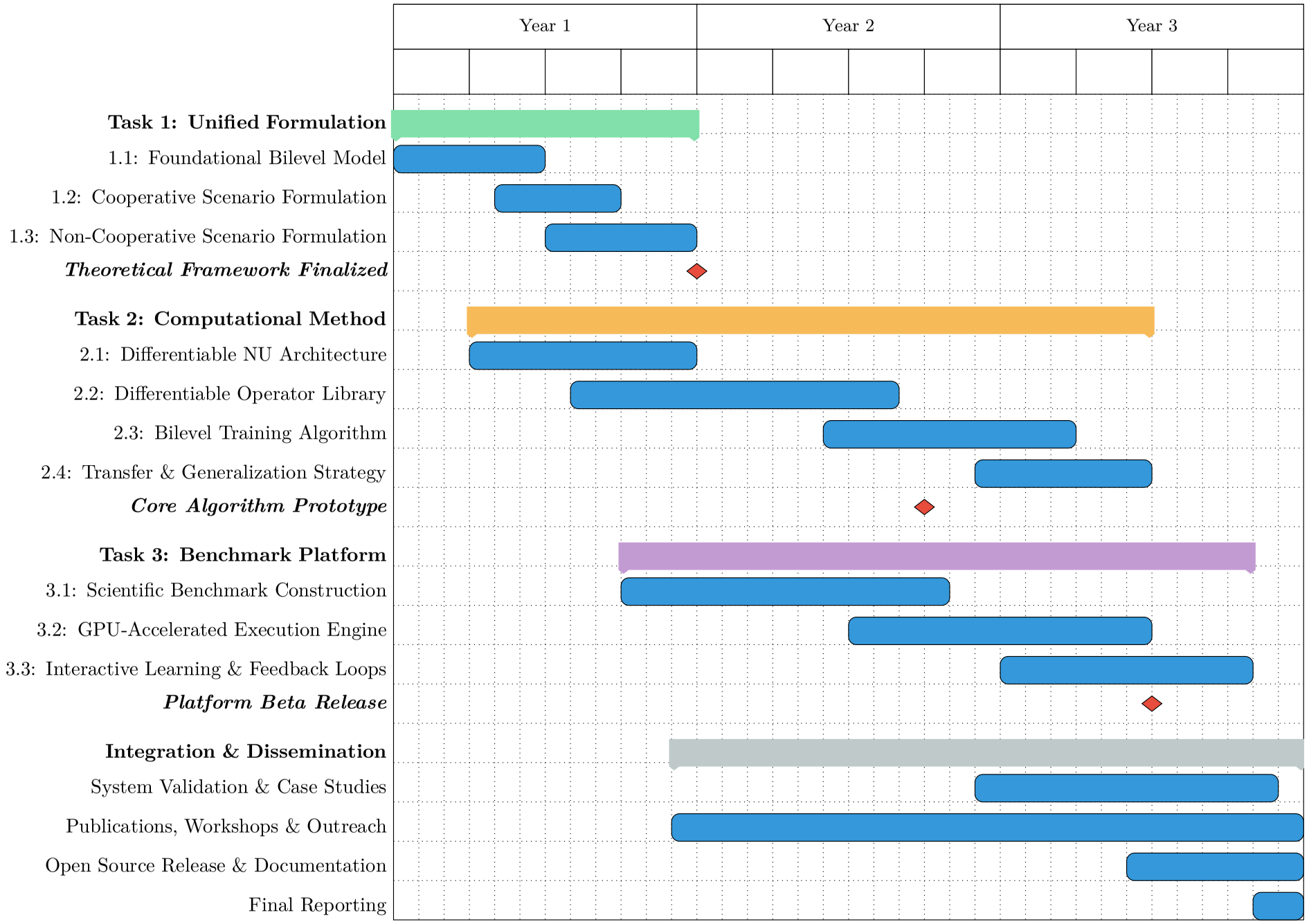
which fuses gradient aggregation and projection into a single kernel, reducing memory transfer overhead.

Task 3.3: Interactive Learning & Feedback Loops

To integrate expert knowledge into the optimization cycle, we incorporate human-in-the-loop feedback. The platform provides coordinated visual analytics of Pareto fronts in both objective and decision spaces, together with temporal trajectories, allowing experts to inspect trade-offs and mark promising regions. Feedback is captured as (i) pairwise preferences among candidate solutions and (ii) targeted improvement signals on attributes, which are converted into differentiable penalties:

$$\mathcal{L}_{sl}^{\text{aug}} = \mathcal{L}_{sl} + \lambda_p \mathcal{L}_{\text{pref}} + \lambda_i \mathcal{L}_{\text{imp}}. \tag{10}$$

Here, $\mathcal{L}_{\text{pref}}$ encodes preference comparisons and $\mathcal{L}_{\text{imp}}$ captures improvement directions. This formulation incorporates domain-driven constraints while preserving gradient-based training, enabling continuous refinement of Pareto solutions.


Year 1
Year 2
Year 3
Task 1: Unified Formulation
1.1: Foundational Bilevel Model
1.2: Cooperative Scenario Formulation
1.3: Non-Cooperative Scenario Formulation
Theoretical Framework Finalized
Task 2: Computational Method
2.1: Differentiable NU Architecture
2.2: Differentiable Operator Library
2.3: Bilevel Training Algorithm
2.4: Transfer & Generalization Strategy
Core Algorithm Prototype
Task 3: Benchmark Platform
3.1: Scientific Benchmark Construction
3.2: GPU-Accelerated Execution Engine
3.3: Interactive Learning & Feedback Loops
Platform Beta Release
Integration & Dissemination
System Validation & Case Studies
Publications, Workshops & Outreach
Open Source Release & Documentation
Final Reporting

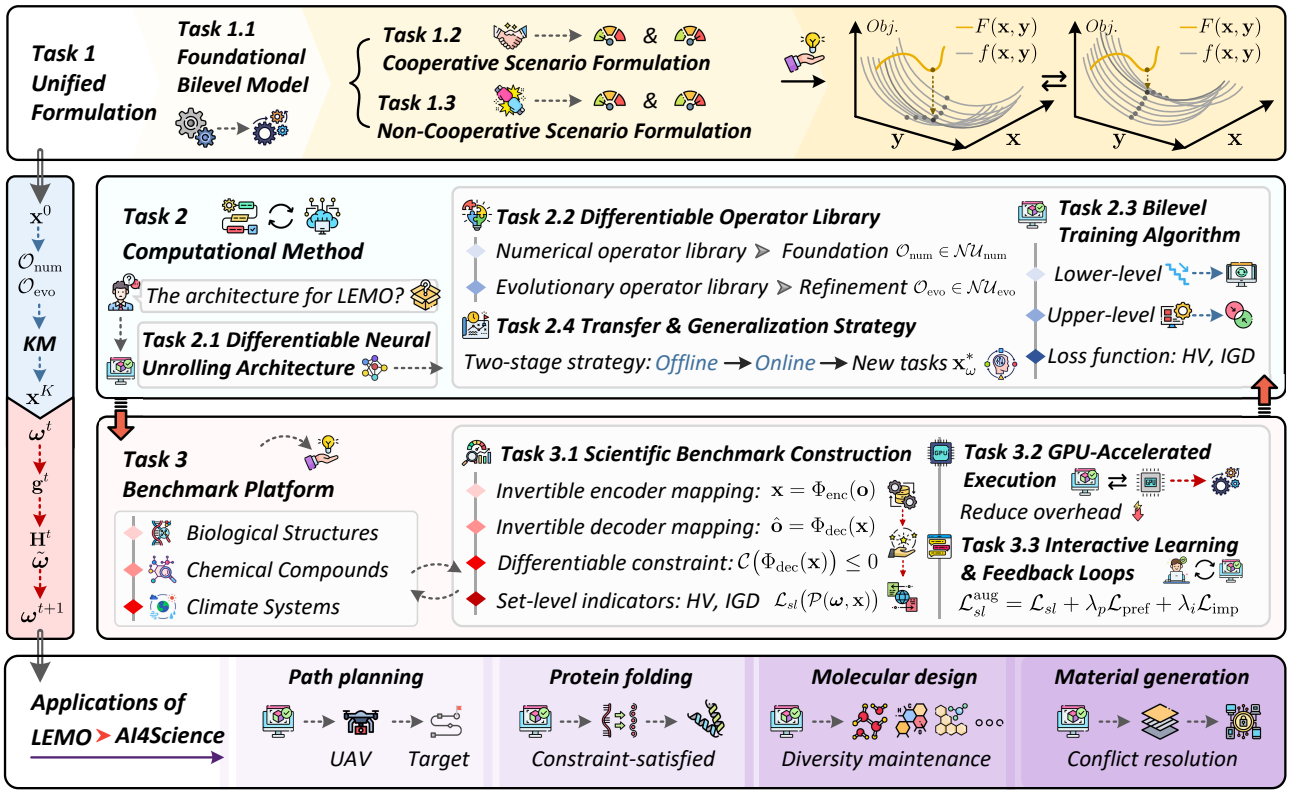


Figure 2: Overall organization of the proposed research framework, consisting of three major tasks: unified formulation (Task 1), computational method (Task 2), and benchmark platform (Task 3), together with their interconnections. Each task is further decomposed into subtasks addressing foundational modeling, modular operator design, bilevel training, transfer and generalization strategies, and validation in representative AI4Science applications.

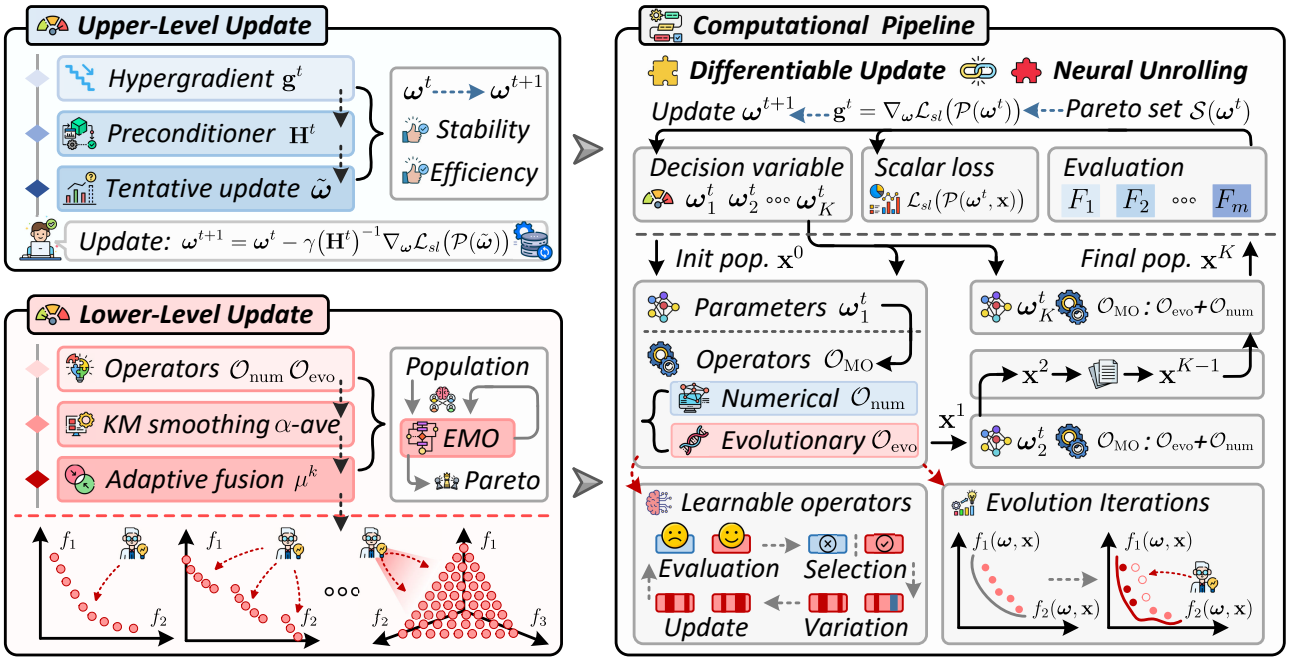


Figure 3: Illustration of the proposed LEMO framework trained in a bilevel manner. The left panel elucidates the hierarchical optimization structure, delineating both upper-level and lower-level update mechanisms. The right panel presents the comprehensive computational pipeline, illustrating the end-to-end workflow and information propagation through the integrated system components.

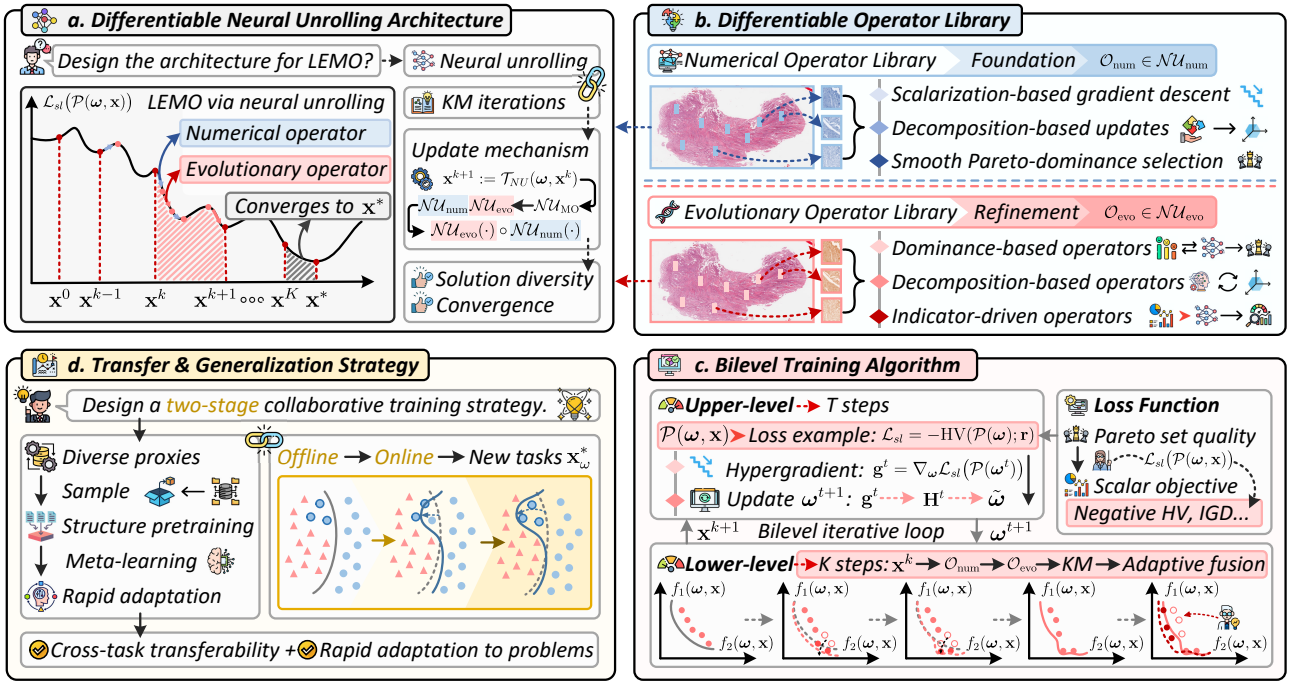


Figure 4: Overview of Task 2, showing the workflow of the differentiable neural unrolling framework: (a) NU architecture that couples numerical and evolutionary operators with KM iterations; (b) modular operator library supporting scalarization, decomposition, dominance, and indicator-driven strategies; (c) bilevel training algorithm integrating adaptive fusion, extragradient updates, and differentiable set-level objectives; (d) two-stage transfer strategy with offline pretraining and online adaptation.

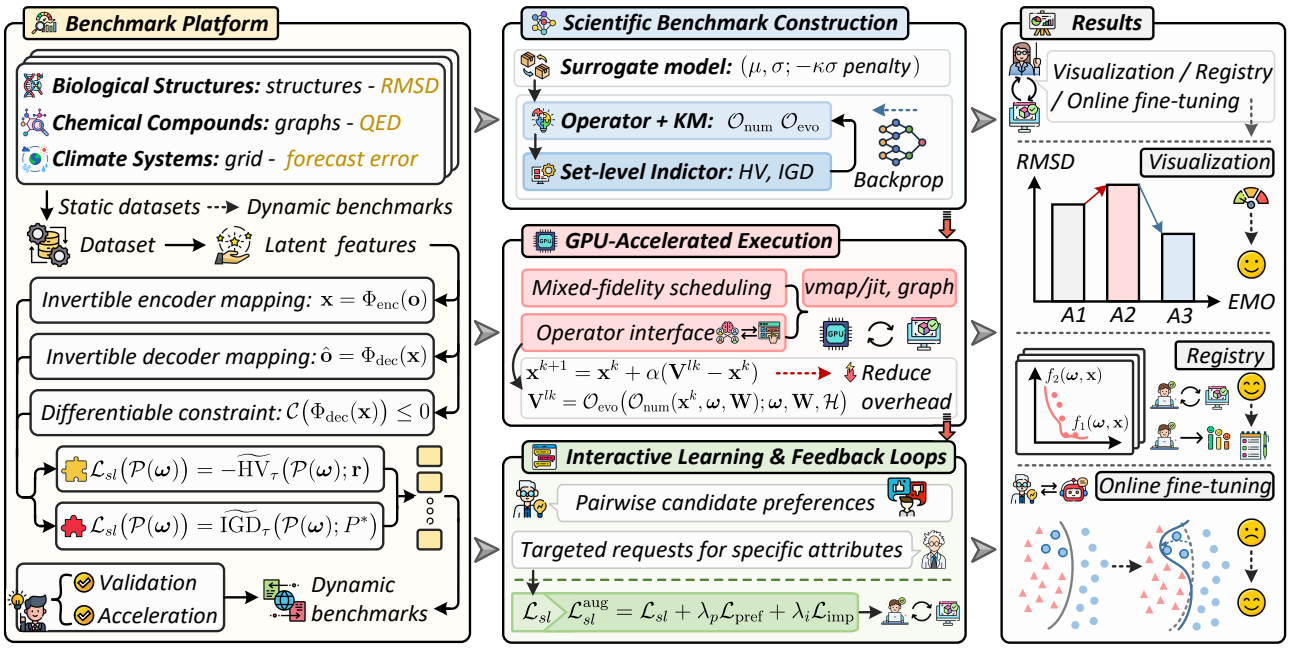


Figure 5: Overview of Task 3, showing the system-level platform for validating and deploying LEMO in AI4Science: (a) construction of differentiable scientific benchmarks via latent encoding, constraints, and surrogate models; (b) integration of neural unrolling with set-level indicators such as HV and IGD; (c) GPU-accelerated execution with tensorized operators and mixed-fidelity scheduling; (d) human-in-the-loop feedback encoded as preferences or improvement requests.

Pathways to Impact Statement

Artificial Intelligence for Science (AI4Science) is rapidly transforming discovery in molecular design, drug development, and materials engineering. Landmark achievements, such as AlphaFold's success in predicting protein structures, illustrate how AI can tackle problems long considered intractable, shrinking research cycles from years to minutes. This momentum is reinforced by substantial investment and institutional support. The AI drug discovery market, for example, is projected to grow from about USD 1.4 billion in 2023 to nearly USD 6.9 billion by 2029, with longer-term forecasts exceeding USD 16 billion by 2034. Broader life-sciences AI markets show a similar trajectory, with estimates rising from about USD 9.8 billion in 2024 to USD 33.5 billion by 2029. Meanwhile, strategic national efforts such as the U.S. Materials Genome Initiative underscore the recognition that algorithmic methods are foundational to future innovation.

Within this accelerating context, multi-objective optimization remains a central methodological challenge. Real-world problems demand trade-offs among conflicting objectives: drug candidates must balance potency, selectivity, toxicity, and manufacturability; materials must satisfy strength, weight, cost, and durability. Evolutionary Multi-objective Optimization (EMO) has long provided a flexible paradigm, yet conventional methods rely heavily on hand-crafted, non-differentiable operators such as non-dominated sorting, crowding distance, and heuristic variation schemes. These choices undermine gradient-based learning, limit adaptability, and prevent integration with differentiable AI frameworks. As problems scale to many objectives and high dimensions, EMO algorithms face convergence–diversity trade-offs and suffer from the curse of dimensionality. Constrained and black-box settings pose additional difficulties, where constraint handling is often heuristic and unreliable in the presence of uncertainty or noise. The widening methodological gap between AI4Science requirements and the limitations of traditional optimization motivates this project.

In response, we propose Learnable Evolutionary Multi-objective Optimization (LEMO), a class of algorithms that combine the strengths of evolutionary search with gradient-based refinement. The project will deliver: a unified mathematical formulation that coherently integrates evolutionary and gradient components within a loss-driven model; a neural unrolling computational framework that enables end-to-end differentiation through optimization iterations; and an open-source benchmark platform tailored to AI4Science tasks, including molecular objectives, materials constraints, and trajectory planning, with reproducible evaluation pipelines.

In constructing the impact pathway, we distinguish clearly between *research outputs* (e.g., code, datasets, tutorials) and *research impacts* (e.g., verifiable improvements in productivity, cost efficiency, usability). Both the reach (i.e., who benefits across which domains) and the significance (i.e., the extent of improvement or change in practice) will be described in accessible and non-technical terms to clarify why the outcomes matter and how they can be adopted.

(i) Who are the potential beneficiaries of the proposed research in the short (1–3 years), medium (4–10 years), and long term (over 10 years)?

Short-term. Beneficiaries will primarily be researchers and graduate students in multi-objective and black-box optimization, as well as AI4Science method developers. The project will release open-source implementations, curated benchmarks, and reproducible workflows, presenting differentiable operators as modular replacements for classical heuristics. To broaden accessibility, it will include tutorials, example notebooks, and template pipelines. Locally, Hong Kong's InnoHK research clusters and universities will be engaged through co-hosted tutorials and student projects, strengthening talent development and interdisciplinary collaboration. Stakeholders will be identified concretely, for example computational chemists at Hong Kong Science and Technology Parks, robotics groups working on UAV planning, and materials researchers employing differentiable simulators.

Medium-term. As the framework stabilizes, beneficiaries will expand to R&D teams in biotechnology, advanced materials, and engineering. The platform will provide stable APIs, GPU-accelerated execution, and configurable operator libraries. Expected benefits include shorter iteration cycles in

virtual screening, fewer wet-lab trials, and more efficient exploration of design spaces under constraints such as safety, durability, and sustainability. Small and medium-sized enterprises in Hong Kong and the Greater Bay Area that lack extensive compute resources will be able to adopt LEMO through containerized deployments and cloud-ready templates. Engagement will evolve from outreach to co-development through pilot projects integrated into real workflows.

Long-term. In the longer term, beneficiaries will include the broader innovation ecosystem, such as public agencies shaping research and industrial strategies. By providing learnable and differentiable mechanisms for multi-objective optimization, LEMO will enable future applications requiring continuous adaptation, including personalized medicine and resilient energy systems. For Hong Kong, strengthening algorithmic infrastructure complements investments in data and hardware, contributing to a sustainable role in the global AI4Science landscape.

(ii) How will the potential beneficiaries benefit? What will be the demonstrable/measurable benefits beyond academia?

The project defines measurable indicators that go beyond academic outputs. In AI4Science case studies, success will be tracked by time and cost savings relevant to users. Examples include reduction in compute hours needed to meet property thresholds in drug-like molecule design, increase in feasible materials candidates per thousand simulations relative to current baselines, and improvement in robustness and success rate for UAV or robotic planning under dynamic conditions. SMART targets will be adopted where feasible, such as a twenty to thirty percent gain in candidate identification efficiency in a partner materials project under fixed budgets. Open-source adoption will be monitored through downloads, forks, user contributions, and citations in patents or technical reports. These indicators demonstrate how outputs translate into real benefits for users.

(iii) What will be done during and/or after the project to increase the likelihood of achieving the identified benefits and reaching the identified beneficiaries?

The impact plan follows a layered pathway from communication to co-production. Communication will include policy briefs, summaries for non-technical stakeholders, newsletters, and media-ready outputs. Engagement will be promoted through lectures, workshops, and showcases aligned with milestones, with tailored sessions for InnoHK centers and universities. Collaboration will involve an industry advisory group meeting twice yearly and two pilot projects from year two, one in molecular screening and one in engineering design such as UAV planning or energy scheduling, both co-designed with partners and documented as case studies. Commercialization readiness will be explored with technology transfer offices, balancing lightweight licensing with open-source release. Skills and capacity will be built through training, graduate modules, and example curricula linking LEMO concepts to practice. Activities will be aligned with project milestones and budgeted for staff, events, travel, and outputs, while monitoring will track indicators such as trained users, pilot projects, workflow adoption, and measurable reductions in trial-and-error effort.

(iv) What are the potential risks that might adversely affect the achievement of the stated impact, and how will such risks be mitigated?

Risks include adoption barriers, computational scalability, sustainability, and privacy compliance. Non-experts may find the system complex, and this will be addressed with high-level APIs, step-by-step notebooks, a graphical prototype, and one-click domain templates. Large design spaces may strain throughput even with GPUs, and this risk will be mitigated by mixed-fidelity scheduling, principled approximations, and continuous kernel optimization. Maintaining the platform beyond the project may be demanding, and mitigation will include early cultivation of a contributor community, modular governance, and integration into subsequent projects or institutional platforms. Sensitive applications such as biomedical or genomic data may face regulatory constraints, and mitigation will involve optional privacy-preserving modes such as federated learning and differential privacy together with clear compliance documentation.

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8. Declaration of Time Commitment, Grant Record and Related Research Work of Investigator(s)

(a) Declaration of time commitments[1]:

(i) Number of on-going individual research projects[2] currently held by the PI in any capacity (e.g. PI, Co-I, Collaborator, etc.): 1

(ii) Number of on-going collaborative research projects[3] currently held by the PI in any capacity (e.g. PC, Co-PI, Co-I, Collaborator, etc.): 0

If the number declared in (i) exceeds three or the number declared in (ii) exceeds two, please provide the following information:

(iii) Overall amount of time spent on research by the PI in percentage terms:

(iv) Justifications on how the PI would be able to take up this additional research project if funded while maintaining the research quality:

(b) Please complete the table below covering:

(i) completed research work funded from all sources (irrespective of whether from UGC/RGC) in the past five years;

(ii) on-going research work funded from all sources (irrespective of whether from UGC/RGC);

(iii) proposals pending funding approval (irrespective of whether submitted to UGC/RGC); and

(iv) any other work related to this research project outside the 5 year period covered above and irrespective of whether it was / is funded by the UGC or the RGC. Please state what distinguishes this project from that work.

---

[1] Research projects funded from all sources (irrespective of whether from UGC/RGC) should be included, except those funded under the joint research schemes and the fellowship schemes administered by the RGC. Projects for which completion reports have yet to be submitted by the application deadline of the current GRF exercise should also be counted.

[2] Examples include research projects funded under the General Research Fund (GRF), Early Career Scheme (ECS), etc. Individual research projects funded from sources other than UGC / RGC should also be included. For details, please refer to GRF2.

[3] Examples include research projects funded under the Areas of Excellence Scheme (AoE), Theme-based Research Scheme (TRS), Collaborative Research Fund (CRF) and Research Impact Fund (RIF). Collaborative research projects funded from sources other than UGC / RGC should also be included. For details, please refer to GRF2.

Complete the first table for the PI then any Co-Is should be detailed in the second table listed in the order listed above.

[1] Research projects funded from all sources (irrespective of whether from UGC/RGC) should be included, except those funded under the joint research schemes and the fellowship schemes administered by the RGC. Projects for which completion reports have yet to be submitted by the application deadline of the current GRF exercise should also be counted.

[2] Examples include research projects funded under the General Research Fund (GRF), Early Career Scheme (ECS), etc. Individual research projects funded from sources other than UGC / RGC should also be included. For details, please refer to GRF2.

[3] Examples include research projects funded under the Areas of Excellence Scheme (AoE), Theme-based Research Scheme (TRS), Collaborative Research Fund (CRF) and Research Impact Fund (RIF). Collaborative research projects funded from sources other than UGC / RGC should also be included. For details, please refer to GRF2.

Completed

TC2023081405 Role:PI 01 Sep 23 - 01 Mar 25 Non RGC/UGC Funding
6

Funding Source(s) (Amount): Huawei Central Research Institute, China($1,950,000)

Project / Work Title:

Sparse and Scientific Computing Performance Optimization

Project / Work Objective:

Focuses on performance optimization for sparse and scientific computing through co-design of algorithms and high-performance computing architectures.

No. of Hours Per Week Spent by the PI *: 0

Related to the current application: N.A.

FA2019061021 Role:PI 01 Jan 20 - 01 Dec 21 Non RGC/UGC Funding

Funding Source(s) (Amount): Huawei Hisilicon, China($1,395,000)

Project / Work Title:

Model Miniaturization for AI Standards and Hisilicon Chips via Evolutionary Algorithms

Project / Work Objective:

Develops evolutionary model compression and optimization algorithms for AI standardization and efficient deployment on Hisilicon chips.

No. of Hours Per Week Spent by the PI *: 0

Related to the current application: N.A.

61906081 Role:PI 01 Jan 20 - 31 Dec 22 Non RGC/UGC Funding

Funding Source(s) (Amount): National Natural Science Foundation of China, Young Scientists Fund, Category C($251,000)

Project / Work Title:

Modular Deep Neural Architecture Search Based on Evolutionary Multi-objective Optimization

Project / Work Objective:

Investigates modular deep neural architecture search using evolutionary multi-objective optimization to enhance scalability and interpretability of deep models.

No. of Hours Per Week Spent by the PI *: 0

Related to the current application: N.A.

Role:PI 01 Jan 20 - 31 Dec 23 Non RGC/UGC Funding

Funding Source(s) (Amount): Ministry of Industry and Information Technology, China($4,149,000)

Project / Work Title:

Deep Learning Based Aerofoil Design

Project / Work Objective:

Develops deep learning-based aerodynamic design and optimization technologies for aerofoils in civil aircraft applications.

No. of Hours Per Week Spent by the PI *: 0

Related to the current application: N.A.

On-going

2024B1515020019 Role:PI 01 Jan 24 - 31 Dec 27 Non RGC/UGC Funding

Funding Source(s) (Amount): Provincial Fund for Distinguished Young Scholar, Guangdong, China($1,090,000)

Project / Work Title:

Computational Intelligence Based Black-box Optimization and Its Applications

Project / Work Objective:

Develops intelligent black-box optimization frameworks integrating evolutionary computation and learning-based models for large-scale industrial applications.

No. of Hours Per Week Spent by the PI *: 8

Related to the current application: Related

Clarifications on related project:

Although both projects are related to optimization, the research focuses and methodologies are distinct. The ongoing project develops computational intelligence frameworks for black-box optimization, emphasizing algorithmic generality and applications to real-world industrial problems. The present proposal concentrates on developing a learnable and differentiable evolutionary multi-objective optimization framework through neural unrolling and bilevel learning. The two projects are complementary rather than overlapping, differing in research objectives, technical approaches, manpower, and funding sources.

*** Please provide the most updated information.**

If the declared work is related to the current project, please state what distinguishes this project from that work.

On-going

Name of Investigator(s) : Prof TAN, Kay Chen

15229824 Role:PI 01 Jan 25 - 31 Dec 27

Funding Source(s) (Amount): GRF($1,298,709)

Project / Work Title:

Towards Efficient Large Model Training via Learning from Small Models

Related to the current application: N.A.

Name of Investigator(s) : Prof TAN, Kay Chen

15215623 Role:PI 01 Jan 24 - 31 Dec 26

Funding Source(s) (Amount): GRF($1,535,774)

Project / Work Title:

Towards Lightweight Optimization for Complex Problems

Related to the current application: N.A.

Name of Investigator(s) : Prof TAN, Kay Chen

15218622 Role:PI 01 Jan 23 - 31 Dec 25

Funding Source(s) (Amount): GRF($1,168,126)

Project / Work Title:

Advancing Multi-task Designs for Large-scale Multi-objective Optimization

Related to the current application: Related

Clarifications on related project:

Although both projects concern multi-objective optimization, the research focuses are different. The ongoing project investigates multitask design and knowledge transfer mechanisms for large-scale multi-objective problems. The present proposal concentrates on developing a learnable and differentiable evolutionary multi-objective optimization framework through neural unrolling and bilevel learning. The two studies are complementary in scope and methodology, with no overlap in objectives, manpower, or funding.

Name of Investigator(s) : Prof TAN, Kay Chen

11211521 Role:PI 01 Jan 22 - 31 Dec 24

Funding Source(s) (Amount): GRF($1,093,580)

Project / Work Title:

Towards Scalable Multi-Objective Bilevel Optimization: Foundations, Methodologies and Applications

Related to the current application: N.A.

Name of Investigator(s) : Prof Yang, Ming

15307625 Role:PI 01 Jan 26 - 31 Dec 28

Funding Source(s) (Amount): GRF($890,587)

Project / Work Title:

High-throughput design of room-temperature two-dimensional magnetic heterostructures for spintronic devices

Related to the current application: N.A.

Name of Investigator(s) : Prof Yang, Ming

15307124 Role:PI 01 Jan 25 - 31 Dec 27

Funding Source(s) (Amount): GRF($910,742)

Project / Work Title:

Rational design of high-performance interface between high-k dielectrics and two-dimensional semiconductors

Related to the current application: N.A.

Name of Investigator(s) : Prof Yang, Ming

25301523 Role:PI 01 Jan 24 - 31 Dec 26

Funding Source(s) (Amount): ECS($754,378)

Project / Work Title:

Data-driven development of high-k dielectrics for high-performance two-dimensional electronics

Related to the current application: N.A.

Pending

Name of Investigator(s) : Prof TAN, Kay Chen

15235626 Role:PI -

Funding Source(s) (Amount): GRF($1,600,000)

Project / Work Title:

Toward Agentic Intelligence in Evolutionary Transfer Optimization for Complex Problem Solving

Related to the current application: N.A.

RGC Ref No. 15225126

Name of Investigator(s) : Prof Yang, Ming

15306426 Role:PI -

Funding Source(s) (Amount): GRF($1,187,760)

Project / Work Title:

Rational design of rippled two-dimensional semiconductors for high-performance electronics

Related to the current application: N.A.

RGC Ref No. 15225126

PART III: PROJECT FUNDING and RESOURCES

9. Cost and Justification

(a) Estimated cost and resource implications:
[Detailed justifications should be given in order to support the request for each item below]
(a maximum of 500 words for each box)

	Year 1 ($)	Year 2 ($)	Year 3 ($)	Year 4 ($)	Year 5 ($)	Total ($)
(A) One-line Vote Items						
(i) Supporting Staff Costs						
[please read Section 9(A)(i) of the Explanatory Notes GRF2 carefully]						
Types						
Monthly salary x Nos. x Months						
Research Postgraduate Students						$707,760
19,660 * 1 * 12	235,920					
19,660 * 1 * 12		235,920				
19,660 * 1 * 12			235,920			
Research Assistant						$432,000
18,000 * 1 * 12	216,000					
18,000 * 1 * 12		216,000				

Justification:

The proposed project encompasses multiple research components, including neural representation learning, computational model construction, data-driven analysis, neuromorphic audio-visual perception system design, and system-level validation. To ensure smooth progress across these tasks, one full-time Research Assistant will be employed for a duration of 24 months. The RA will provide support in model implementation, experimental execution, data processing, and system integration. In parallel, one research postgraduate student will be trained throughout the 36-month project period, focusing primarily on theoretical modeling and foundational algorithmic studies.

(ii) Equipment Expenses
[please itemize and provide quotations for each item costing over $200,000]

Justification:

Quotation Provided: **Yes** ☐ **No** ☑

(iii) Outsourcing Expenses of Research Work Outside Hong Kong
[please itemize your cost estimation with justification and provide quotations for work costing over $200,000; and provide detailed justification of sample sizes and costs for surveys conducted outside Hong Kong.]

Justification:

Quotation Provided: **Yes** ☐ **No** ☑

(iv) General Expenses
[please itemize and provide quotations for services/purchase costing over $200,000; and provide detailed justification of sample sizes and costs for surveys conducted in Hong Kong.]

Publishing Charges	16,666	16,666	16,666	0	0	49,998

Justification:

The publication expenses will include fees associated with both conference proceedings and journal articles, covering open-access publication costs as well as additional page charges for extended manuscripts.

Quotation Provided: **Yes** ☐ **No** ☑

(v) Conference Expenses

For Attending Workshops and Conferences	16,666	16,666	16,666	0	0	49,998

Justification:

Participation in leading international conferences will enable the research team to disseminate findings more broadly, engage in scholarly exchange, and obtain constructive feedback from the global research community.

Sub-total for (A) (One-line Vote Items): **$** 1,239,756

(B) Earmarked Items

(vi) Costs for Employment of Relief Teacher
[see Enclosure III for individual research and Enclosure V for relief support under Humanities and Social Science Panel]

Rank

Per course rate of relief teacher: x No. of course to be relieved

Justification:

PI's Current Average Teaching Load: Total 0 courses per academic year [UGC-funded programmes only]

(vii) Expenses of Research Experience for Undergraduate Student
(see Enclosure VI for Provision of Research Experience for Undergraduate Students)

Justification:

(viii) High-performance Computing Services Expenses (HPCS)

Total: 0

Justifications including explanation on how the item(s) is/are related to the research activities and essential to deliver the research results, and description on how the budget is arithmetically determined, etc.

Quotation Provided: **Yes** ☐ **No** ☑

(viii)(b) **Enhanced Earmarked Funding to meet exceptionally high demand for HPCS [For proposal requesting for funding exceeding $100,000 in (viii)]**

☐ **By clicking this box, I declare that there are genuine research needs to apply for the enhanced earmarked funding for HPCS. I also understand that I am obliged to provide further justifications to support this request.**

Further justification: including, but not limited to, explanations on whether alternative means to obtain the research results have been explored, whether more than one quotation has been obtained (if the services required are not exclusively provided by a single service provider), alternatives to lower the required HPCS cost and how could the enhanced earmarked fund be gainfully deployed for the research activities, etc.

(ix) Research-related Software Licence /Dataset
[Please itemize and provide quotations for each item]

Justification:

Sub-total for (B) (Earmarked Items):	**$**	0
(x) Total cost of the project (A) + (B)	**$**	1,239,756

(C) Deduction Items

Less :

(xi) University's funding for provision of research experience for undergraduate student	**$**	0

(xii) Other research funds secured from other sources	**$**	0
Sub-total for (C) (Deduction Items) :	**$**	0
(xiii) Amount requested in this application : (A) + (B) - (C)	**$**	1,239,756

(D) Academic Research related to Public Policy Developments

(xiv) Percentage of the total cost of the proposal related to public policy developments ((A) + (B))
[see Enclosure VII for Support for Academic Research relating to Public Policy Developments] 0%

(b) Declaration on the Equipment Procurement:

☑ **(i) No procurement of equipment is required**

OR

☐ **(ii) I declare that the equipment indicated in Section 9(a)(A)(ii) above is not available in the university**

OR

☐ **(iii) I declare that all or some of the equipment (please provide details in the following text box) indicated in Section 9(a)(A)(ii) above is available in the university but cannot be used by this project for the following reasons (a maximum of 500 words)**

Reasons : (a maximum of 500 words)

(c) Declaration on employment of relief teacher:

☑ **(i) No relief teacher is required**

OR

☐ **(ii) I declare that I currently do not hold any grant for employment of relief teacher of any on-going project under UGC/RGC funding schemes**

OR

☐ **(iii) I declare that I hold grant for employment of relief teacher of**

the following on-going project(s) under UGC/RGC funding schemes (excluding Humanities and Social Sciences Prestigious Fellowship Scheme (HSSPFS))

(d) Declaration on high-performance computing services:

☑ (i) No procurement of high-performance computing services is required

OR

☐ (ii) I declare that the high-performance computing services indicated in Section 9(a)(B)(viii) above is not available in the university

OR

☐ (iii) I declare that all or some of the high-performance computing services (please provide details in the following text box) indicated in Section 9(a)(B)(viii) above is available in the university but cannot be used by me in view of the following reasons (a maximum of 500 words)

Reasons : (a maximum of 500 words)

(e) Declaration on the research-related software licence / dataset:

☑ (i) No procurement of research-related software licence / dataset is required

OR

☐ (ii) I declare that the research-related software licence / dataset indicated in Section 9(a)(B)(ix) above is not available in the university

OR

☐ (iii) I declare that all or some of the research-related software licence / dataset (please provide details in the following text box) indicated in Section 9(a)(B)(ix) above is available in the university

but cannot be used by me in view of the following reasons (a maximum of 500 words)

Reasons : (a maximum of 500 words)

10. Existing Facilities and Major Equipment available for this Research Project: (a maximum of 400 words)

Attached 1 page(s) as follows. (I confirm that the number of words in the uploaded file does not exceed 400 words)

All collaborating parties, including the PI and Co-Is from The Hong Kong Polytechnic University and Queen Mary University of London, as well as supporting partners from the A*STAR Institute of High Performance Computing, the University of Electronic Science and Technology of China, and the China Pharmaceutical University, have confirmed their commitment to providing the necessary computing resources, laboratory facilities, and research datasets required for the proposed project. The following major facilities will be available to all team members for algorithm development, benchmarking, and validation.

(1) The Hong Kong Polytechnic University
The PI's laboratory is equipped with a high-performance computing environment comprising 40 NVIDIA RTX 3090 GPUs (24 GB each) and 32 NVIDIA RTX A6000 GPUs (48 GB each), supported by a 64 TB SSD data storage system. These resources will be dedicated to large-scale model training and multi-objective optimization experiments. The university's central computing facility provides additional HPC nodes, high-speed network infrastructure, and licensed research software to support neural architecture development and data-intensive tasks.

(2) Queen Mary University of London
The Co-I, Dr. Yao Lu, will provide access to state-of-the-art materials and chemistry research facilities, including advanced characterization instruments (SEM, TEM, XPS), nanomaterial fabrication tools, and wet-chemistry laboratories. These facilities will be used to construct and validate AI4Science benchmark datasets in functional materials and surface engineering, enabling experimental feedback for algorithmic evaluation.

(3) A*STAR Institute of High Performance Computing
The collaboration with Dr. Liangli Zhen will leverage IHPC's large-scale supercomputing infrastructure, including multi-GPU clusters, high-memory compute nodes, and distributed AI training environments. These facilities will be used to evaluate the scalability, generalization, and robustness of the proposed optimization algorithms under real-world computational workloads.

(4) University of Electronic Science and Technology of China
Prof. Xianfu Wang's research group will provide access to neuromorphic computing systems and experimental hardware facilities, including device-level testing setups, circuit integration platforms, and simulation environments for electronic thin films and microsystems. These facilities will support exploratory experiments linking the learnable EMO framework to intelligent hardware implementations.

(5) China Pharmaceutical University
Prof. Yu Sun's Institute of Intelligent Pharmacy will provide molecular modeling platforms, and proprietary compound databases. These resources will be used to test and validate the proposed optimization framework in AI-driven drug discovery, multi-objective pharmacological modeling, and molecular design. The collaboration will ensure that the learnable EMO framework can be evaluated across diverse scientific domains and data modalities.

RGC Ref No. 15225126

11. Funds Secured or to be Secured

(a) Other research funds already secured for this research proposal:
[This amount will be deducted from the total cost of the project in Part III Section 9 above.]

Source	Amount ($)

(b) Other research funds to be or are being sought for this research proposal.
[If funds under this item are secured, the amount of the GRF to be awarded may be reduced]:

Source	Amount ($)

PART IV: ETHICS AND OTHER APPROVALS

12. Research Ethics / Safety Approval

(i) I confirm that the research proposal ☐ **involves /** ☑ **does not involve human subjects.**

(ii) I confirm that the research proposal ☐ **involves /** ☑ **does not involve the study of artefacts.**

(iii) Please tick in the appropriate boxes to confirm if approval for the respective ethics and/or safety issues is required and has been / is being obtained from the PI's university. PIs are encouraged to seek necessary approval (except for human research ethics (clinical)) before application deadlines.

	Approval not required	Approval being sought	Approval obtained
(1) Animal research ethics	☑	☐	☐
(2) Biological safety	☑	☐	☐
(3) Ionizing radiation safety	☑	☐	☐
(4) Non-ionizing radiation safety	☑	☐	☐
(5) Chemical safety	☑	☐	☐
(6) Human research ethics (non clinical)	☑	☐	☐
(7) Artefacts research ethics	☑	☐	☐

	Approval not required	Approval being sought	Approval obtained	Approval will be sought if funded
(8) Human research ethics (clinical)	☑	☐	☐	☐

(iv) If approval is required by other authorities, please indicate *below* the names of the authorities and the prospects of obtaining such approval. If not applicable, please put down "N.A.".

N.A.

13. Access to Government/ Official/ Private Data and Records

(i) Is access to Government / official / private data and records critical to the research project?

☐ **Yes**

☑ **No**

If approval is required, please indicate below the names of the agency(ies) of obtaining such approval.

(ii) Please tick in the appropriate boxes to confirm if approval for access to the related data/records has been / is being obtained from the relevant agency(ies). If approval has been obtained, please provide evidence.

List of agency(ies)	Approval not required	Approval being sought	Approval obtained

14. Project Data

(i) Is the proposed project likely to generate data set(s) of retention value?

Yes ☐ **No** ☑

If yes, please describe the nature, quantity and potential use of the data set(s) in future.

Nil

(ii) Are you willing to make the data set(s) available to others for reference twelve months after the publication of research results or the completion of this proposed project?

Yes ☐ **No** ☐

I understand that the RGC will release the completion report to the public and only considers data archiving requests after the completion of the RGC-funded project. The RGC has full discretion in funding the archiving requests. Data sets archived with RGC funds will require users to acknowledge the originator and the RGC. The originator will also be provided with copies of all publications derived from the use of the data.

I undertake to include in the project completion report the URL links to the university's repository or the publishers' websites so that the public could have quick and easy access to the manuscripts or journal articles. I will also consider to include in the research completion report the data repository where research data of the project could be accessed and shared, where appropriate.

I undertake that upon acceptance of a paper for publication,

(i) I will check whether the publisher already allows (A) full open access to the publisher's version, or (B) my depositing a copy of the paper (either the publisher's version or the final accepted manuscript after peer-review) in the university's repository for open access;

(ii) if both (i) (A) and (B) are not allowed, I will request the publisher to allow me to place either version in my university's repository for restricted access immediately upon publication or after an embargo period of up to twelve months if required by the publisher; and

(iii) subject to the publisher's agreement on (i) or (ii) above, I will deposit a copy of the publication in my university's repository as early as possible but no later than six months after publication or the embargo period, if any.

PART V: ADDITIONAL ATTACHMENTS

Appendix 1: Education Plan

Appendix 3: Letters of Collaboration with other research team(s) or university(ies)
Support Letters from Dr. Liangli Zhen, Prof. Xianfu Wang, and Prof. Yu Sun.

Appendix 5: CV Submission for Applications for Funding for Post-doctoral Fellows and / or Other Supporting Documents
CV of Dr. Jiaxin Gao (Post-doctoral Researcher, Supporting staff; no salary requested from GRF)

Education Plan

The proposed project advances methodological research in Learnable Evolutionary Multi-objective Optimization (LEMO) while also emphasizing education and training. A key objective is to cultivate a new generation of researchers and practitioners who can bridge theory and practice in optimization and AI4Science. In the context of Hong Kong and the Greater Bay Area, this aligns with the region's strategic emphasis on developing innovation talent in artificial intelligence, data science, and computational engineering. The education plan consists of four interconnected components: student training, course teaching, outreach and dissemination, and industry collaboration.

Student Training. The multidisciplinary scope of LEMO, which spans optimization, machine learning, bilevel modeling, and tensorized computation, offers a rich environment for student development. Through structured research projects and close supervision, students will acquire both theoretical foundations and practical skills. PhD students will focus on innovation in algorithm design and contribute to high-quality publications, while Master's students will emphasize translating new methods into applied solutions for multi-objective optimization tasks. Undergraduate students will progressively build a grounding in optimization theory, programming, and reproducibility, with opportunities to participate in senior projects and early-stage research. This tiered training structure will help prepare students for both academic careers and technical roles in industry.

Course Teaching. The project outcomes will be incorporated into graduate seminars and advanced undergraduate electives. New course modules will cover the theoretical basis of bilevel optimization and the practical aspects of implementing learnable EMO algorithms. Students will work on benchmark experiments, coding assignments, and group projects that highlight reproducibility and critical evaluation. Graduate students will be encouraged to take leadership roles in research projects, while undergraduate students will gain early exposure to applied optimization through capstone projects using the LEMO framework. By embedding project outcomes into the curriculum, students will gain hands-on experience with state-of-the-art methods that complement their theoretical knowledge.

Outreach and Dissemination. Active engagement in dissemination will be a central element of the education plan. Students will be encouraged to present their work at international conferences in optimization, artificial intelligence, and evolutionary computation, allowing them to gain exposure to the global research community. The PI will organize workshops and tutorials focusing on unified modeling, neural unrolling, and differentiable operators, offering platforms for students to practice presentation and collaboration. Locally, outreach activities will also target undergraduate students through seminars on optimization and AI4Science case studies, fostering early interest in research and awareness of cross-disciplinary applications.

Industry Collaboration. Industry partnerships will be integrated into the training plan. PhD students will have opportunities to participate in research internships with industrial partners engaged in AI-driven optimization and engineering applications. These internships will expose students to real-world multi-objective challenges and resource-constrained environments, reinforcing the relevance of their academic training. Industry partners will also contribute to workshops and seminars, providing domain-specific feedback and highlighting practical needs. Such interactions will help students align their skills with industry requirements and prepare them for successful careers across academic, industrial, and entrepreneurial pathways.

Agency for Science, Technology and Research
SINGAPORE

Agency for Science, Technology and Research (A*STAR)
1 Fusionopolis Way
#16-16, Connexis North Tower
Singapore 138632

25 September 2025

Dear Prof. Cheng,

At the Institute of High Performance Computing (IHPC), under A*STAR, we are committed to advancing computational science and developing intelligent systems that can address critical challenges across domains such as healthcare, manufacturing, and national infrastructure. A key part of this mission is to explore methods that integrate machine learning with evolutionary optimization, since such approaches open new possibilities for adaptive design, automated decision-making, and large-scale simulation.

In this context, I am very pleased to express my strong support for your grant proposal entitled “Towards Learnable Evolutionary Multi-objective Optimization via Neural Unrolling”. Your project brings together rigorous optimization theory with learnable algorithmic design, creating a framework that is both scalable and adaptive. We believe this work will have far-reaching implications for evolutionary multi-objective optimization and at the same time provide practical tools that are directly relevant to many of our ongoing research programs.

The ability to apply learnable optimization methods to highly complex systems, where reliability, adaptability, and efficiency must all be achieved simultaneously, would directly benefit several areas of IHPC research. We are confident that your project will generate outcomes of both academic importance and practical value, and we look forward to the possibility of working together to explore these applications further.

Sincerely yours,

Liangli Zhen

Liangli Zhen, PhD
Group Manager & Senior Scientist
Department of Computing & Intelligence
Institute of High Performance Computing
Agency for Science, Technology and Research (A*STAR), Singapore

State Key Laboratory of Electronic Thin Films and Integrated Devices
University of Electronic Science and Technology of China
Xiyuan Ave. 2006, 611731, West Hi-Tech Zone | Chengdu, Sichuan, China

Prof. Ran Cheng
Department of Data Science and Artificial Intelligence
The Hong Kong Polytechnic University
[24 Sep 2025]

Letter of Support for Prof. Ran Cheng

Dear Prof. Cheng,

As the leader of Neuromorphic Computing Devices and Systems Group at the State Key Laboratory of Electronic Thin Films and Integrated Devices, I am pleased to express my strong support for your RGC proposal entitled *"Towards Learnable Evolutionary Multi-objective Optimization via Neural Unrolling"*.

Our laboratory has been committed to advancing frontier research in electronic thin films, integrated devices, and microsystems, with neuromorphic computing devices and systems as one of our key strategic directions. In this area, the challenges of high-dimensional design and the limitations of traditional computing architectures highlight the urgent need for scalable and adaptive optimization frameworks.

Your proposal directly addresses this need by introducing an innovative and rigorous optimization methodology that is highly relevant to our ongoing research. We believe it will not only advance evolutionary optimization as a discipline but also provide valuable tools for intelligent hardware design, neuromorphic system development, and unconventional computing, which are areas that are central to our laboratory's mission.

We also see strong opportunities for collaboration. The proposed framework could significantly accelerate optimization tasks essential for materials discovery, thin film integration, and neuromorphic circuit design. Such synergy would further strengthen cross-disciplinary efforts to combine advanced optimization algorithms with microelectronics and integrated microsystems research.

For these reasons, I am pleased to endorse your proposal and confident that it will generate substantial academic and technological impact. I look forward to its successful outcomes and to future collaborations between your group and ours.

Best regards,

Xiangfu Wang

State Key Laboratory of Electronic Thin Films and Integrated Devices
University of Electronic Science and Technology of China

中国药科大学
CHINA PHARMACEUTICAL UNIVERSITY

#639 Longmian Avenuel Jiangning District
Nanjing,211198,P.R.China
Tel: 0086-25-83271423
Fax:0086-25-83213611
www.cpu.edu.cn

Dear Prof. Cheng,

As Executive Dean of the Institute of Intelligent Pharmacy at China Pharmaceutical University, I am pleased to offer my strong support for your GRF proposal entitled *"Towards Learnable Evolutionary Multi-objective Optimization via Neural Unrolling"*.

My professional background spans more than two decades at Intel China, where I was responsible for advancing high performance computing, AI, and their applications in healthcare and life sciences. Since joining China Pharmaceutical University in 2024 as a Distinguished Research Professor, I have focused on establishing the Institute of Intelligent Pharmacy, dedicated to integrating AI with the full pipeline of drug discovery.

Your proposed research is highly relevant to the mission of our institute. Particularly, it addresses one of the central challenges in intelligent drug discovery, i.e., the efficient exploration of vast and nonlinear search spaces in molecular design and pharmacological modeling. The framework you propose is not only methodologically rigorous but also highly adaptable, and it aligns closely with the urgent need for scalable AI-driven optimization in pharmaceutical research.

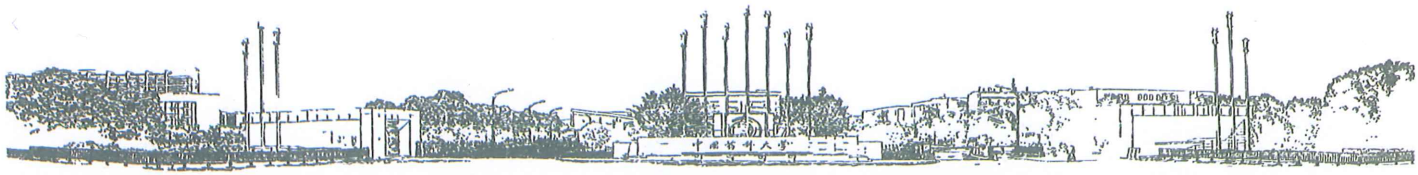
Applying your optimization methods to problems in drug target identification, compound screening, and multi-objective evaluation of efficacy and safety could greatly enhance the efficiency and reliability of intelligent drug discovery workflows. Such outcomes would not only advance academic research but also create significant value for translational medicine and industrial innovation.

Therefore, I am delighted to endorse your proposal and confident that it will generate substantial academic and technological impact. I look forward to future collaborations with you.

Sincerely,

Yu Sun.
30/09/2025

Prof. Yu Sun
Executive Dean
Institute of Intelligent Pharmacy
China Pharmaceutical University



GAO Jiaxin, Ph. D. | Postdoctoral Fellow
Department of Data Science and Artificial Intelligence
Hong Kong Polytechnic University
Phone: (+86) 18340833067
Email: jiaxinn.gao@gmail.com
ORCID: 0000-0002-0023-1269

Education

- 2020.09 - 2024.09 Ph.D. in Software Engineering, Dalian University of Technology, China.
- 2018.09 - 2020.06 M.S. in Computational Mathematics, Dalian University of Technology, China.
- 2014.09 - 2018.06 B.S. in Applied Mathematics, Dalian University of Technology, China.

Employment

- 2025.04 - Present Postdoctoral Researcher, Hong Kong Polytechnic University, Hong Kong.

Research Interest

My research focuses on constructing reliable and efficient intelligent optimization algorithms for core tasks at the intersection of machine learning and multimedia. Bilevel optimization serves as a key mathematical tool for handling hierarchical and coupled sub-problems, modeling complex deep learning paradigms, and solving high-dimensional vision applications, showing great potential in areas like large model learning. My work addresses theoretical and computational challenges in hierarchical optimization and complex coupled vision tasks, developing tailored efficient algorithms for applications such as meta-learning, neural architecture search , and hierarchical low-level vision tasks, with growing interest in scalable multi-objective frameworks for AI4Science applications. To date, I have published 15 SCI/EI papers, including journals and conferences such as *IEEE TPAMI, IEEE TNNLS, CVPR, IJCAI, ACM MM, AAAI, and Pattern Recognition*. My work has received over 390 citations (Google Scholar). Additionally, I have contributed to 3 patent applications.

Publication List – Recent Publications

1) **Jiaxin Gao**, Ziyu Yue, Yaohua Liu, Sihan Xie, Xin Fan, Risheng Liu. A Dual-Stream-Modulated Learning Framework for Illuminating and Super-Resolving Ultra-Dark Images. *IEEE Transactions on Neural Networks and Learning Systems* 35, 4567-4580, 2024.
2) **Jiaxin Gao**, Yaohua Liu. Enhancing Images with Coupled Low-Resolution and Ultra-Dark Degradations: A Tri-level Learning Framework. *Proceedings of the ACM International Conference on Multimedia* 880-891, 2024.
3) **Jiaxin Gao**, Yaohua Liu, Ziyu Yue, Xin Fan, Risheng Liu. Collaborative Brightening and Amplification of Low-Light Imagery Via Bi-Level Adversarial Learning. *Pattern Recognition* 152, 110-125, 2024.
4) **Jiaxin Gao**, Xiaokun Liu, Xin Fan, Risheng Liu. Learning adaptive hyper-guidance via proxy-based bilevel optimization for image enhancement. *The Visual Computer* 39, 2455-2468, 2023.
5) Risheng Liu, **Jiaxin Gao**, Jin Zhang, Deyu Meng, Zhouchen Lin. Investigating Bi-Level Optimization for Learning and Vision from a Unified Perspective: A Survey and Beyond. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 104, 105-128, 2022.
6) Risheng Liu, **Jiaxin Gao**, Xuan Liu, Xin Fan. Learning with Constraint Learning: New Perspective, Solution Strategy and Various Applications. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 118, 205-225, 2024.

7) Ziyue Yue, **Jiaxin Gao**, Zhixun Su. Unveiling Details in the Dark: Simultaneous Brightening and Zooming for Low-Light Image Enhancement. *Proceedings of the AAAI Conference on Artificial Intelligence* 38, 1205-1215, 2024.
8) Yaohua Liu, **Jiaxin Gao**, Xuan Liu, Xianghao Jiao, Xin Fan, Risheng Liu. Advancing Generalized Transfer Attack with Initialization Derived Bilevel Optimization and Dynamic Sequence Truncation. *Proceedings of the International Joint Conference on Artificial Intelligence*, 1205-1215, 2024.
9) Xin Fan, Xiaolin Wang, **Jiaxin Gao**, Jia Wang, Zhongxuan Luo, Risheng Liu. Bi-level Learning of Task-Specific Decoders for Joint Registration and One-Shot Medical Image Segmentation. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 880-891, 2024.
10) Xiaofeng Liu, **Jiaxin Gao**, Yaohua Liu, Nenggan Zheng, Risheng Liu. Motion-Scenario Decoupling for Rat-Aware Video Position Prediction: Strategy and Benchmark. *Proceedings of the International Conference on Image and Graphics*, 345-356, 2023.
11) Xianghao Jiao, Yaohua Liu, **Jiaxin Gao**, Zhu Liu, Xin Fan, Risheng Liu. PEARL: Preprocessing Enhanced Adversarial Robust Learning of Image Deraining for Semantic Segmentation. *Proceedings of the ACM International Conference on Multimedia*, 650-661, 2023.
12) Sihan Xie, Peiming Li, **Jiaxin Gao**, Ziyu Yue, Xin Fan, Risheng Liu. Breaking the water dilemma: Transmission-guided bilevel adaptive learning for underwater imagery. *Neurocomputing* 550, 126-135, 2023.
13) Bo Fu, Yunyun Jiang, Di Wang, **Jiaxin Gao**, Cong Wang, Ximing Li. Uncertainty-aware Sparse Transformer Network for Single Image Deraindrop. *IEEE Transactions on Instrumentation and Measurement* 73, 1-12, 2024.

Patents

- A robust motion prediction method for bio-robots integrating adversarial training. Risheng Liu, Xiaofeng Liu, Yaohua Liu, **Jiaxin Gao**. *PCT/CN2023/117444*, 2023.
- A method for predicting future actions of bio-robots by integrating visual images and neural signals. Risheng Liu, Xiaofeng Liu, Yaohua Liu, **Jiaxin Gao**. *PCT/CN2023/117453,* 2023.
- A single-sample image segmentation method based on bilevel coupled learning. Xin Fan, Risheng Liu, Zhongxuan Luo, **Jiaxin Gao**. *PCT/SG*2023114042244*,* 2023.

Awards

- 2015, 2016, National Inspirational Scholarship of Dalian University of Technology
- 2018, 2019, 2020, First-Class Academic Scholarship of School of Mathematical Sciences
- 2018, Second Prize in Liaoning Provincial Post-Graduate Mathematical Contest, Northeastern University

Service

- Conference PC Member: CVPR, ACM MM, AAAI, ICCV, ICLR, NeurIPS, IJCAI, AISTATS, WACV
- Journal Reviewer: PR, TCSVT, TNNLS, TEVC, TVC, Neurocomputing, Springer Nature

THE HONG KONG POLYTECHNIC UNIVERSITY
香港理工大學

Prof. Kay Chen Tan
Department of Data Science and AI
The Hong Kong Polytechnic University
Email: kaychen.tan@polyu.edu.hk
6 Oct 2025

Prof. Ran Cheng
Department of Data Science and Artificial Intelligence
The Hong Kong Polytechnic University

RE: Letter of Collaboration on GRF Fund

Dear Prof. Cheng,

This is to confirm that I agree to participate in the Hong Kong General Research Fund (GRF) proposal entitled "Towards Learnable Evolutionary Multi-Objective Optimization via Neural Unrolling". I will actively contribute as a Co-Investigator and allocate the necessary resources to support its implementation. I am fully committed to fulfilling the responsibilities outlined in the project proposal in a timely and proactive manner.

KC Tan

Yours sincerely,
Prof. Kay Chen Tan

THE HONG KONG POLYTECHNIC UNIVERSITY
香港理工大學

Letter of Collaboration

Prof. Ming Yang
Department of Applied Physics
The Hong Kong Polytechnic University
Email: mingyang@polyu.edu.hk

6 October 2025

To:
Prof. Ran Cheng
Department of Data Science and Artificial Intelligence
The Hong Kong Polytechnic University

Subject: Confirmation of Collaboration for GRF Project "Towards Learnable Evolutionary Multi-Objective Optimization via Neural Unrolling"

Dear Prof. Cheng,

I am writing to confirm my participation as a Co-Investigator in the above GRF proposal. I will contribute my expertise in materials physics, high-throughput modeling, and AI4Science validation, and will assist in constructing and evaluating the scientific benchmarks related to materials discovery.

I am pleased to support this collaboration and will devote the necessary effort and resources to ensure the project's successful completion.

With kind regards,



Ming Yang

Dr. Yao Lu
Senior Lecturer, Department of Chemistry
Queen Mary University of London (QMUL), UK
Email: yao.lu@qmul.ac.uk
5 Oct 2025

Dr. Ran Cheng
Department of Data Science and Artificial Intelligence
The Hong Kong Polytechnic University

RE: Letter of Collaboration for GRF Proposal "Towards Learnable Evolutionary Multi-Objective Optimization via Neural Unrolling"

Dear Dr. Cheng,

This is to confirm that I agree to participate in the Hong Kong General Research Fund (GRF) proposal titled "*Towards Learnable Evolutionary Multi-Objective Optimization via Neural Unrolling*" as a Co-Investigator.

I will contribute domain expertise in materials chemistry and support benchmark/data curation and validation relevant to the project. I am committed to allocating the necessary time and resources and to fulfilling the responsibilities described in the proposal in a timely and proactive manner.

Yours sincerely,

Yao Lu



RGC Ref No. 15225126

PART VI: UNDERTAKING OF THE PI

By submitting the application, I undertake that the research activity(ies) / expenditure(s) of the project if funded will be carefully monitored for its / their compliance with applicable laws, health and safety guidelines and ethical standards, including but not limited to the Law of the People's Republic of China on Safeguarding National Security in the Hong Kong Special Administrative Region.